Sheffield Hallam University | Faculty of Business Technology and Engineering

MSc Dissertation Report

Evaluating the Security of the NHS Spine's Personal Demographics Service (PDS) and Its Integration with the Parkinson's KinetiGraph Wearable Device: An Analysis of Data Handling Practices and Potential Threats

A dissertation submitted in partial fulfilment of the requirements of Sheffield Hallam University for the degree of Master of Science in Cyber Security.

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ABSTRACT

This study critically examines the security implications of integrating the Parkinson's KinetiGraph (PKG) wearable device with the NHS Spine's Personal Demographics Service (PDS). As healthcare systems increasingly adopt wearable technologies, understanding and mitigating associated security risks becomes crucial. This research aims to evaluate current security measures, identify potential vulnerabilities, and propose recommendations for enhancing overall system security.

A qualitative case study approach was employed, utilizing document analysis and literature review to investigate the NHS Spine infrastructure, PDS functionality, and PKG integration. The study focused on three key areas: existing security measures in NHS systems, the impact of PKG integration on data security, and recommendations for mitigating identified risks.

Findings reveal that while the NHS Spine and PDS employ robust security measures, including strong authentication and encryption protocols, the integration of wearable devices introduces new vulnerabilities. Key risks identified include potential unauthorized access, data interception, and challenges in maintaining data integrity across integrated systems. The study also highlights the complexities of managing patient consent and data privacy in the context of continuous health monitoring.

Based on these findings, the research proposes a comprehensive set of recommendations, including implementing enhanced authentication protocols, securing data transmission pathways, and establishing rigorous API management practices. A phased approach to security enhancement is suggested, prioritizing critical measures while addressing medium-priority concerns.

This study contributes to bridging the gap between wearable technology in healthcare and security in healthcare IT systems. It provides a foundation for future research in this rapidly evolving field and offers practical insights for healthcare organizations and technology providers. The findings underscore the need for a dynamic, multi-layered approach to healthcare data security in the era of wearable technology integration.

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1 INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The National Health Service (NHS) in the United Kingdom has been at the forefront of digital transformation in healthcare, with the NHS Spine playing a crucial role in connecting various healthcare systems and services (Gibbs, 2018). The NHS Spine, which handles up to 47 million messages every day, is a critical infrastructure that enables secure communication between 28,000 NHS IT systems across 21,000 care units (NHS Digital, 2024j). NHS Spine supports several key services that are essential for healthcare delivery as shown in the figure below.

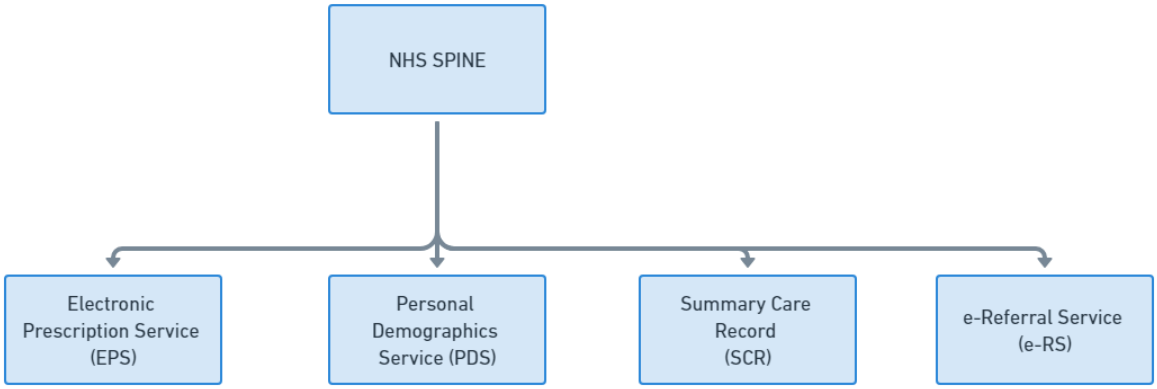

Figure 1: NHS Spine and Core Services

1. Personal Demographics Service (PDS): A core component of the NHS Spine, serves as a centralized database containing demographic information for over 90 million patients (NHS Digital, 2024g).
2. Summary Care Record (SCR): A national database that provides healthcare professionals with quick access to crucial patient information such as current medication for over 65 million patients (NHS Digital, 2024f).
3. Electronic Prescription Service (EPS): This enables the electronic transfer of approximately 2 million prescriptions daily between prescribers and pharmacies (NHS Digital, 2024c).
4. NHS e-Referral Service (e-RS): This allows patients to easily select their hospital and appointments with a specialist (NHS Digital, 2024m).

The NHS is integrating wearable devices, such as the Parkinson's KinetiGraph (PKG), into its digital infrastructure to collect real-time patient data. The NHS Spine, especially the Personal

Demographics Service (PDS), facilitates the connection of this data to the appropriate patient records, potentially revolutionizing the management of chronic conditions like Parkinson's disease.

The PKG, a wrist-worn device, continuously monitors patient movements, providing clinicians with comprehensive symptom reports, enabling data-driven decisions and individualized treatments (Dementech, 2022). NICE endorses these devices, which could potentially benefit up to 120,000 individuals in the UK (NHS England, 2022).

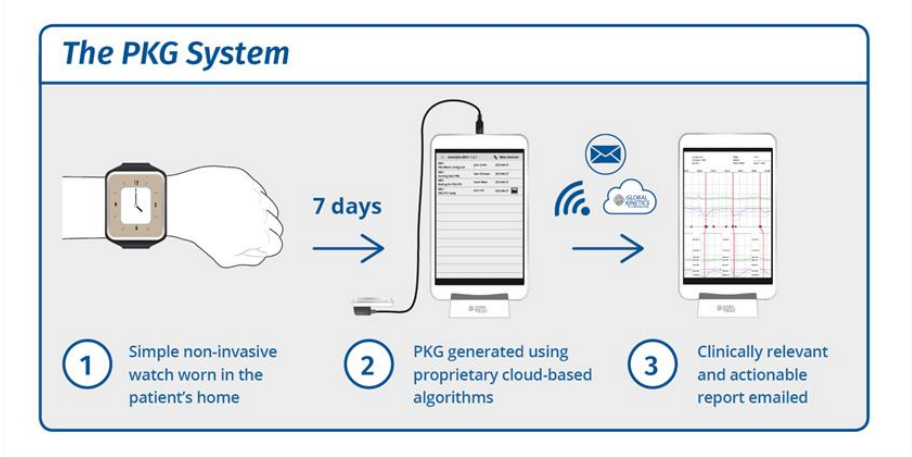

Figure 2: PKG System. Retrieved from (Shake It Up, 2019)

Wearable sensors have become popular for monitoring and managing PD symptoms. Advanced systems like PKG and PDMonitor offer continuous tracking of bradykinesia, dyskinesia, tremor, gait freezing, and postural instability (Reichmann et al., 2023). These systems often feature smartphone apps for patients to log medication intake and communicate with physicians, facilitating remote monitoring and treatment adjustments (Bendig et al., 2022; Reichmann et al., 2023).

However, increased use of wearable technology for health monitoring raises concerns about data security and privacy, as these devices collect, store, and share personal health information (Mone & Shakhlo, 2023). The General Data Protection Regulation (GDPR) offers a framework for protecting privacy rights, but ensuring compliance and addressing vulnerabilities remains challenging (Mone & Shakhlo, 2023).

1.2 PROBLEM STATEMENT

Integrating wearable devices with the NHS Spine offers notable benefits but also poses challenges, particularly concerning data privacy and security. A study found many users worried about the collection, storage, and sharing of their health data (Mone & Shakhlo, 2023).

The 2017 WannaCry ransomware attack affected over 230,000 computers globally, disrupting numerous NHS facilities (Mohurle & Patil, 2017). In 2020, the SolarWinds supply chain attack compromised several NHS organizations (NHS Digital, 2021a), and in 2024, the Synnovis cyber-attack led to the publication of stolen patient data, delaying over 6,000 outpatient appointments and 1,400 elective procedures (Synnovis, 2024).

Stakeholders, including healthcare providers, patients, and regulatory bodies, are concerned about implementing secure systems while benefiting from digital transformation. Dr. Amir Mehrkar, interim Chief Medical Officer at NHS Digital, emphasizes the importance of interoperability and FHIR standards for data sharing and patient care improvement (Hoeksma, 2019).

1.2.1 SIGNIFICANCE OF STUDY

Given these significant security challenges and their potential impacts, it is crucial to examine the integration of wearable devices with the NHS Spine's PDS. As wearable technology in healthcare becomes more prevalent, ensuring the security and confidentiality of patient data is essential. By analyzing current security measures and data handling practices within the PDS and its integration with PKG this study aims to identify vulnerabilities and propose recommendations for enhancing system security.

The findings can contribute to developing more secure and trustworthy digital healthcare ecosystems, benefiting patients, healthcare providers, and the NHS. Addressing security concerns associated with wearable device integration can facilitate the safe and effective adoption of these technologies, ultimately improving patient care and outcomes.

Moreover, the insights from this research may have broader implications for integrating wearable devices with other healthcare systems beyond the NHS, advancing secure and privacy-preserving digital health solutions globally.

1.2.2 WEARABLE DEVICES AND PARKINSON'S DISEASE MANAGEMENT

Kinesia 360, KinesiaU, PDMonitor, Personal KinetiGraph (PKG), and STAT-ON are conditionally recommended wearable devices options for remote monitoring of Parkinson's disease to inform clinical decision-making, as stipulated by the National Institute for Health and Care Excellence (NICE, 2023).

Figure 3: KinesiaU Device. Retrieved from (KinesiaU, 2024)

Figure 4: PDMonitor. Retrieved from (PD Neurotechnology, 2024).

Figure 5: Parkinson's KinetiGraph (PKG). Retrieved from (PKG, 2016)

Figure 6: STAT-ON System. Retrieved from (SINAPSI, 2020).

To highlight the potential benefits and challenges of integrating wearable devices with the NHS Spine’s PDS, this study will focus on a specific case of Parkinson's disease management using the PKG.

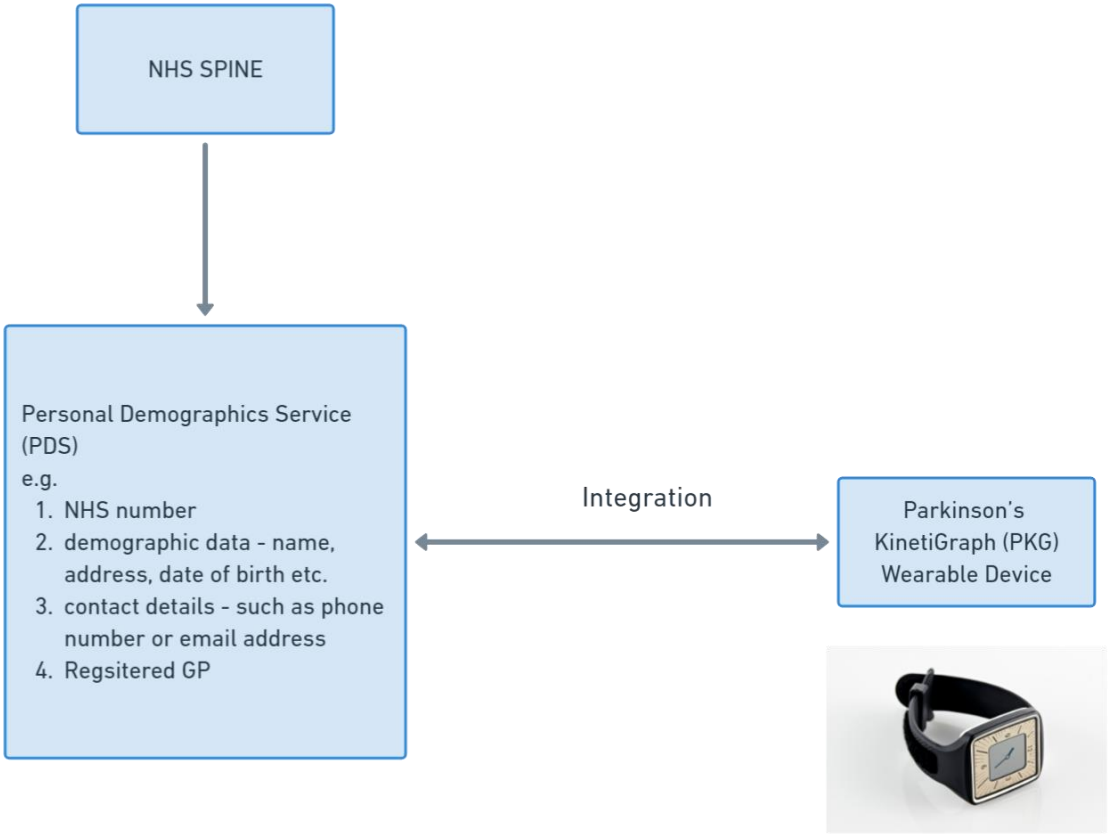

Figure 7: NHS Spine's PDS-PKG System Integration

A pilot study conducted by Bendig et al. (2022) on a multimodal telemedical intervention for Parkinson's disease patients, involving wearable sensors, video visits, and a smartphone app. The study revealed that most elderly patients could effectively use digital technologies, but success hinged on close support from a coordinator and physician. In the U.S., the FDA has approved several wearable devices for remote monitoring, such as the Empatica Embrace2, a wrist-worn device for detecting seizures in epilepsy patients, which has improved clinical practice and patient outcomes (Empatica, 2024).

These cases highlight the potential of wearable devices to enhance care and cut costs. The IQVIA Institute for Human Data Science estimated that wearable devices could save the NHS up to £7.1 billion annually by 2025, emphasizing the need to address security and privacy issues to ensure benefits without compromising data (IQVIA, 2021).

1.2.3 DATA SHARING AND CONSENT PROCESSES

Data sharing and consent processes are essential when integrating wearable devices with PDS. The NHS adheres to strict protocols for obtaining patient consent and data sharing, in accordance with the UK Data Protection Act 2018 and the EU GDPR (ICO, 2022).

To protect patient data, the NHS implements measures like Role-Based Access Control, NHS Smartcards, NHS Care Identity Service 2, and the Data Security and Protection Toolkit (NHS Digital 2023d, NHS England 2023b, NHS Digital 2024k, NHS Digital 2023b). Furthermore, the NHS adheres to security standards and guidelines established by the National Data Guardian to ensure robust data protection (NDG, 2024).

Mone and Shakhlo (2023) highlighted privacy risks of wearable devices, such as unauthorized access to health data, sale or disclosure of personal information without user consent, and potential discrimination based on health data. They stressed the need for strong security measures, greater user control, and regulation of data use in wearable devices. Gandarilla (2018) also underscored the importance of robust authentication, role-based access controls, and strong password policies to protect sensitive health data.

1.3 RESEARCH AIM AND OBJECTIVES

Given the complex landscape of wearable device integration, data security concerns, and regulatory requirements, this research aims to critically examine NHS Spine's Personal Demographics Service (PDS) and its integration with the Parkinson's KinetiGraph wearable device, focusing on the handling of sensitive patient health information, and to identify potential security threats, confidentiality, integrity, and availability issues, ultimately proposing high-level recommendations to enhance the security of these systems. This study seeks to contribute to the development of a more secure and trustworthy digital healthcare ecosystem that benefits both patients and healthcare providers.

The objectives of this research are:

1. Provide a concise overview of NHS Spine Core services, investigate the NHS Spine's Personal Demographics Service (PDS), and the key security measures employed to protect patient data within these core services.

2. Analyse the integration of the Parkinson's KinetiGraph wearable device with the PDS, evaluating the security implications and data handling practices specifically related to this integration.
3. Identify critical potential security threats within the PDS and its integration with the Parkinson's KinetiGraph, and propose recommendations to mitigate risks to patient data confidentiality, integrity, and availability

1.4 RESEARCH QUESTIONS

1. What are the current security measures and data handling practices employed by the NHS Spine's Personal Demographics Service (PDS) to protect sensitive patient health information, and how effective are these measures in mitigating potential security threats?
2. How does the integration of the Parkinson's KinetiGraph wearable device with the PDS impact data security, and what are the specific vulnerabilities and risks associated with this integration in terms of patient data confidentiality, integrity, and availability?
3. Based on the identified security threats and vulnerabilities within the PDS and its integration with the Parkinson's KinetiGraph, what are the key recommendations for enhancing the overall security of these systems, and how can these recommendations be prioritized and implemented?

1.5 OVERVIEW OF THE RESEARCH STRUCTURE AND OUTLINE

This dissertation examines the security of the NHS Spine's PDS and its integration with the PKG wearable device. It focuses on data-handling practices, security threats, and enhancement recommendations. The study begins with an introduction and research question, followed by a literature review. The methodology section details the research methods, data collection techniques, and ethical considerations. Findings will be discussed in relation to existing literature, concluding with recommendations and areas for further study. The dissertation ends with a critical reflection on research limitations and potential improvements for the industry and organization.

2 LITERATURE REVIEW

2.1 INTRODUCTION

This Literature Review outlines research relevant to assessing the security of the NHS Spine's PDS and its integration with the PKG wearable device. It explores themes like NHS Spine infrastructure and security, healthcare wearables, data governance and privacy, Parkinson's disease management, and cybersecurity strategies. The review identifies gaps in the literature and provides context and justification for the proposed study.

2.2 NHS SPINE INFRASTRUCTURE AND SECURITY

Integrating wearable devices, such as PKG, with the NHS Spine infrastructure, particularly the Personal Demographics Service (PDS), poses significant security challenges. As healthcare systems digitize, understanding these challenges is essential to safeguarding patient data privacy and system integrity. This review explores recent literature on NHS Spine infrastructure and security, emphasizing the implications of integrating wearable devices.

Gibbs (2018) offered a thorough overview of the NHS Spine network infrastructure and its cybersecurity threats using a descriptive analysis approach. The study details the Spine network's components and highlights potential vulnerabilities such as end-user device compromise, insider threats, and issues with the HTTP-based messaging system. While this study excels in identifying specific vulnerabilities, it lacks concrete recommendations for mitigating these threats. Nonetheless, Gibbs' work is crucial for understanding the security landscape of the NHS Spine and PDS, laying the groundwork for evaluating risks associated with wearable device integration.

Building upon this foundation, Zhang et al. (2022) examined IT infrastructure challenges in the NHS, conducting a literature review and analysis to identify major barriers such as financial constraints, leadership issues, and insufficient clinical engagement. Although the study clearly outlines these problems and discusses them thoroughly, it lacks specific recommendations, offering only broad suggestions. Nevertheless, the work is relevant for understanding the organizational and systemic factors affecting the secure integration of wearable devices with NHS systems like Spine’s PDS.

Shifting focus to a historical perspective, Honeyman et al. (2016) reviewed the NHS digital health agenda and implementation plans, examining key government and NHS strategy

documents, progress, and associated barriers and opportunities. Although the paper comprehensively addresses national digitization plans and assesses challenges, it could have further explored critical issues like infrastructure, interoperability, and data security. Nonetheless, their discussion on data sharing and information governance challenges is important to the security and confidentiality concerns of integrating wearable devices with NHS systems such as the Spine's PDS.

Expanding on these findings, Rushdi and Kamal (2012) thoroughly examined management information systems (MIS) within the NHS, detailing the organizational structure, modernization initiatives, and IT's impact on service delivery. They provided practical future recommendations, although the paper's methodology and evidence were occasionally lacking. Despite these limitations, the paper provides essential context on NHS information systems and management approaches, vital for understanding the NHS Spine's PDS environment.

2.3 WEARABLE DEVICES IN HEALTHCARE

The integration of wearable devices with healthcare information systems, particularly the NHS Spine and Personal Demographics Service (PDS), presents both opportunities and challenges for patient care and data security. This review examines recent literature on wearable devices in healthcare, with a focus on their potential integration with NHS infrastructure and the associated security implications.

Cai et al. (2017) conducted a case-controlled study using smart wrist bracelets to measure daily activity in Parkinson's disease (PD) patients. The study showed that wearable technology could continuously and objectively monitor PD symptoms. However, its small sample size and focus on gross motor function limit its applicability. Despite these limitations, the research offers valuable insights into data collection and handling by wearable devices, essential for understanding the security implications of integrating such technologies with NHS systems.

Building upon this foundation, Wu et al. (2024) conducted a two-week longitudinal observational study to validate commercial wearable devices for circadian rhythm assessment. By comparing consumer device data to reference measures, they demonstrated these technologies' potential for real-world health monitoring. Strengths include concurrent monitoring of multiple physiological parameters and evaluation of data collection stability. However, notable limitations are the absence of a validated at-home melatonin measurement

method and limited participant diversity. This research is crucial for understanding the capabilities and potential security implications of integrating consumer wearables with NHS systems.

Expanding on these findings, Reichmann et al. (2023) comprehensively reviewed wearable technologies for diagnosing and treating Parkinson's disease, focusing on sensor systems like the PKG for continuous symptom monitoring. Their firsthand experience with various systems enhances their analysis, though the absence of detailed quantitative data on accuracy and reliability is a limitation. Despite this, the review is essential for evaluating potential security risks when integrating these devices with NHS infrastructure.

Shifting focus to a related aspect, Henshall et al. (2022) utilized a qualitative method to assess wearable devices and suggest policies for their broader adoption. Their analysis of benefits and practical recommendations offers valuable insights into integrating wearables within healthcare systems. Nonetheless, the absence of quantitative data and limited discussion on potential drawbacks, such as privacy issues, are significant weaknesses. Despite these limitations, their recommendations could guide the evaluation of risks and security measures necessary for integrating wearables with systems like the NHS Spine's PDS.

Taking a different approach, Smuck et al. (2021) analyzed successful implementations of wearable technologies in healthcare by examining digital health programs at two major health systems. Their identification of key success factors, including technology support and data integration, provides valuable insights for integrating wearables into healthcare systems like the NHS Spine's PDS. However, the study's focus on only two health systems may limit its generalizability.

2.4 DATA GOVERNANCE AND PRIVACY

The integration of wearable devices like the PKG with the NHS Spine infrastructure presents significant challenges in data governance, privacy, and security. As healthcare systems increasingly adopt digital technologies, understanding these challenges becomes crucial for maintaining the integrity and confidentiality of patient data while leveraging the benefits of wearable devices.

Jones et al. (2023) offer a comprehensive review of key UK legal and information governance frameworks, including the Data Protection Act 2018, GDPR 2018, and the Health

and Care Act 2022. Using a descriptive approach, they analyze these frameworks and present best practice examples. Although the paper effectively summarizes the complex data governance landscape, its brevity restricts in-depth analysis of each regulation. This research sets the regulatory groundwork for integrating PKG data with the NHS Spine PDS.

Building upon this foundation, Mone and Shakhlo (2023) focused specifically on the use of digital biomarkers in psychiatric research and the associated data protection requirements under the GDPR. The authors conducted a detailed analysis of GDPR articles and principles, providing valuable insights into compliance challenges when working with sensitive personal data collected by digital devices. Although the study's focus on psychiatric research may limit generalizability, it remains highly relevant to the PKG-PDS integration, as the PKG would be considered a digital biomarker.

Shifting focus to public attitudes, Belfrage et al. (2022) conducted a cross-sectional descriptive study examining the relationship between trust in the Swedish healthcare system's handling of electronic health data and attitudes towards various data uses. The study employed a survey methodology to gather data from the general public. Despite limitations such as a low response rate, this research is relevant to understanding potential security threats related to patient trust and data sharing behaviors in the context of integrating wearable devices with NHS systems.

Expanding on these findings, Atkin et al. (2021) assessed public awareness and perceptions of secondary use of health data using a mixed-methods approach. The study's strengths include the co-development of recommendations with stakeholders and its examination of public attitudes towards data opt-out systems in the UK. This research provides valuable insight into public perspectives that should be considered when assessing security risks and measures for the NHS Spine's PDS.

2.5 PARKINSON'S DISEASE MANAGEMENT AND WEARABLES

The integration of wearable devices with healthcare systems presents significant opportunities and challenges, particularly in the context of the NHS Spine infrastructure and its security implications. This review examines recent literature on the use of wearable devices for Parkinson's disease management and their potential integration with NHS systems, focusing on security considerations and data management challenges.

Poplawska-Domaszewicz et al. (2024) demonstrated the feasibility of using the PKG wearable device to detect early morning akinesia in Parkinson's patients by continuously monitoring motor symptoms over 6-7 days, leading to targeted dopaminergic therapies. Although the study's real-world application and potential to guide treatment decisions are strengths, its small sample size and lack of a control group limit generalizability. The research underscores the need for secure and reliable data transmission from wearable devices to clinical systems, crucial for NHS Spine integration.

Building upon this foundation, AlMahadin et al. (2020) employed a qualitative approach to explore perspectives of Parkinson's disease patients and healthcare professionals on assessment methods and wearable devices. Their user-centered methodology, utilizing semi-structured interviews and focus groups, emphasizes the importance of end-user involvement in device design. However, the small sample size limits the broader applicability of findings. This study underscores the necessity for objective data collection using wearable devices, directly relating to data integrity considerations for NHS Spine integration.

Expanding on these findings, Dominey et al. (2020) conducted a three-year study on the routine use of the PKG in Parkinson's disease care at a UK center, involving 166 patients. They assessed clinical utility and patient acceptability, highlighting real-world application and practical implementation challenges. Limitations included evolving methodology and data management issues across multiple users. This research offers insights into integrating wearable devices with healthcare systems, especially concerning NHS Spine security.

2.6 CYBERSECURITY THREATS AND MITIGATION STRATEGIES

Several recent studies have demonstrated the growing threat of cyberattacks on healthcare systems, underscoring the need for robust cybersecurity measures. Gibbs (2018) evaluated potential vulnerabilities in the NHS Spine network, such as insider threats and weaknesses in the HTTP-based messaging system. The study's comprehensive analysis of the infrastructure and identification of specific vulnerabilities provide valuable insights for the proposed research, highlighting the need to assess and address potential security risks in the PKG-PDS integration.

Another study by Iacobucci (2024) reported on a ransomware attack affecting multiple London hospitals, demonstrating the vulnerability of NHS IT infrastructure to cyberattacks

and the potential consequences for patient care. While the study did not take into account the technical aspects of the attack or the NHS's cybersecurity measures in detail, it emphasizes the importance of considering the broader cyber threat landscape when evaluating the security of wearable device integrations.

However, another recent study by Wu et al. (2024) confirmed the effectiveness of commercial wearable devices for circadian rhythm assessment in home environments. While the study does not specifically address data security, it explores the data handling practices of wearable devices linked to healthcare systems, highlighting the benefits and challenges of using commercial sensors for remote patient monitoring. Future research could expand on these insights by analyzing the data security implications of integrating the PKG with the PDS.

Several key areas of cybersecurity threats to healthcare data systems and wearable devices emerge, as shown in Table 1. Although the list covers major threats, it is not exhaustive, given the constantly evolving cybersecurity landscape. These threats are emphasized due to their prevalence in recent literature, relevance to wearable devices, alignment with research objectives (e.g., threats and vulnerabilities within PDS-PKG integration), and their broad impact on healthcare data security.

Threat	Description	Examples/Implications
Data Breaches	Unauthorized access to healthcare systems and wearable devices storing patient data (Boumpa et al., 2022)	- Leak of 61 million fitness tracker records from Apple and Fitbit. - Exposure of personal information (names, birth dates, weight, height, gender, location). - Privacy violations and potential misuse of information. Source: (Boumpa et al., 2022)
Malware and Ransomware Attacks	Malicious software designed to disrupt operations or extort funds.	- Synnovis Attack - Disruption of healthcare services. - Financial losses.

		(Source: Iacobucci, 2024)
Insider Threats	Misuse or exposure of patient information by individuals with authorized access (Cheema et al., 2023)	- Risk to NHS Spine (Gibbs, 2018) - Intentional or unintentional data leaks. - Need for strong access controls and user activity monitoring.
Device Vulnerabilities	Security weaknesses in medical technologies, including electronic health records, wireless infusion pumps, endoscope cameras, and radiology information systems (Mejía-Granda et al., 2023)	- Poor credential management and authentication issues. - Risk of unauthorized access, data corruption, or execution of malicious code. - Recommendations for manufacturers to adopt more robust security practices. (Source: Mejía-Granda et al., 2023)
Unsecured Data Transmission	Transmission of sensitive health data over insecure wireless networks, leaving them vulnerable to eavesdropping, interception, and unauthorized access (Cilliers, 2019; Mejía-Granda et al., 2023)	- Lack of interoperability standards (Canali et al., 2022). - Poor encryption practices and insecure communication channels in medical devices and health software (Kruse et al., 2017).
Compliance Challenges	Difficulties in meeting regulatory requirements like HIPAA and GDPR (Cheema et al., 2023; Berezin and Berezin, 2024)	- Compared to "walking a tightrope" (Said et al., 2024). - Risk of penalties for non-compliance. - Resource allocation issues potentially creating cybersecurity gaps.

Table 1: Common Cybersecurity Threats in Healthcare and Their Implications

This table below outlines some key critical strategies for mitigating cybersecurity risks in healthcare. Each strategy is supported by recent literature and best practices in the field.

Risk Mitigation Strategy	Description
Comprehensive Risk Assessment	The comprehensive risk assessment of PKG-PDS integration should focus on gathering and interpreting information to reduce uncertainty within the complex system. Nathan and Bhandari (2022) emphasize the development of an explanatory formulation that considers various factors' interactions and examines psychological, technical, and environmental elements. Rather than making categorical risk predictions, the assessment should characterize plausible future scenarios and involve relevant stakeholders to understand the risks collaboratively. The goal is to inform decision-making to improve the overall security and resilience of the integrated system while acknowledging inherent uncertainties.
Strong Authentication and Access Control	Implementing strong authentication and access control measures is critical for preventing unauthorized access to sensitive healthcare data in health information systems. Shojaei et al. (2024) recommends using robust authentication mechanisms like multi-factor authentication and enforcing the principle of least privilege. Employing access control models such as role-based access control (RBAC) or attribute-based access control (ABAC) is also crucial. Gupta et al. (2023) stresses the importance of encrypting data both at rest and in transit. Similarly, Mohamed et al. (2022) suggests that access control policies should be granular and context-aware, considering factors like user roles, data sensitivity, and environmental conditions. Regular auditing and monitoring of access attempts is important for detecting potential breaches. Overall, a defence-in-depth approach combining multiple security controls and following security best practices is recommended to adequately safeguard sensitive medical information.
Secure Data Transmission and Encryption	The use of secure data transmission protocols and encryption techniques is essential for safeguarding healthcare data during transit between systems. Goswami et al. (2023) suggests employing hybrid cryptographic models that combine optimization algorithms with encryption techniques to enhance security. For instance, the ASS-JFO-DHEA model integrates Artificial Shuffle Shepherd and Jellyfish optimization algorithms with

	Digital Homomorphic Elgamal Algorithm encryption to provide multi-layered security and enable efficient key management and parallel processing. Narula et al. (2022) emphasize that these advanced encryption techniques ensure data confidentiality, integrity, and privacy during transmission, making them crucial for securing data exchanges in IoT-based healthcare systems.
Incident Response and Disaster Recovery	Developing incident response plans and disaster recovery strategies is crucial for healthcare organizations to minimize cyberattack impact. Argaw et al. (2020) emphasizes the importance of regular testing, exercising, and offline storage of these plans. Waclawski (2015) highlights that resilience-promoting strategies allow quick recovery after incidents. Preparation, including mock exercises and simulations, ensures minimal disruption to patient care and operations. These measures collectively contribute to a robust incident response and disaster recovery framework.
Cybersecurity Training and Awareness	Healthcare organizations need comprehensive cybersecurity training to promote secure practices and reduce human error risks. Frati et al. (2024) report that over 90% of staff lack adequate cybersecurity awareness, underscoring the critical need for effective training programs. McDade (2024) recommends that these programs should cover HIPAA compliance, phishing detection, social media threats, and device security. The training should be designed with engaging, accessible modules and clear reporting mechanisms to ensure maximum effectiveness and retention of critical cybersecurity knowledge among healthcare staff.

Table 2: Key Cybersecurity Risk Mitigation Strategies for Healthcare Systems

COMPARATIVE ANALYSIS OF STUDIES							
Name of Author with Titles	**Scope of Study**	**Methodology**	**Key Findings**	**How is data collected?**	**How is data secured?**	**What device is used to collect data?**	**Agreement with other studies**
Gibbs M. "Evaluating Cyber Threats to the United Kingdom's National Health Service (NHS) Spine Network"	The study encompasses the technical components of the NHS Spine network and the identification of potential security vulnerabilities.	Qualitative review of existing literature and analysis of NHS Spine network.	Identified potential cybersecurity threats to the NHS Spine network, including insider threats and challenges associated with HTTP-based messaging.	Not explicitly discussed, as the focus is on the NHS Spine network infrastructure and cybersecurity threats.	Discusses the importance of data security in the context of the NHS Spine network, mentioning role-based access control, NHS Smartcards, and the challenges	Not applicable, as the study focuses on the NHS Spine network infrastructure and does not involve direct data collection.	Consistent with other studies that highlight the potential cybersecurity threats to healthcare networks and the need for robust security measures.

					associated with HTTP-based messaging.		
AlMahadin et al. "Parkinson's disease: current assessment methods and wearable devices for evaluation of movement disorder motor symptoms – a patient and healthcare professional perspective"	The study focuses on the perspectives of PD patients and healthcare professionals concerning the use of wearable devices for monitoring PD symptoms.	Qualitative study using semi-structured interviews and focus group discussions.	Highlighted the potential benefits of continuous, objective monitoring using wearable devices and the importance of user-centered design in their development and integration.	Data is collected through wearable devices, such as the Parkinson's KinetiGraph, which continuously monitor PD patients' motor symptoms.	Emphasizes the significance of data security and privacy considerations when integrating wearable devices with the PDS, but does not provide specific details on how data is	The study discusses the use of wearable devices, specifically the Parkinson's KinetiGraph, for collecting data on PD patients' motor symptoms.	Aligns with previous research on the benefits of wearable devices for PD symptom monitoring and the importance of user-centered design in their development and integration.

					secured.		
Jones et al. "Navigating data governance associated with real-world data for public benefit: an overview in the UK and future considerations"	Overview of key legal and information governance frameworks in the UK for responsible use, linkage and sharing of real-world health data.	Descriptive overview and discussion of regulations, good practice exemplars, and future considerations.	Understanding the data governance landscape, including key regulations like the DPA 2018, GDPR, common law duty of confidentiality, NHS Act 2006 section 251, COPI Regulations 2002, and Health and Care Act 2022, is crucial for enabling responsible	Not specified as the paper focuses on governance of real-world health data in general.	The importance of robust data governance frameworks, consent mechanisms, anonymisatio n techniques and strict access controls to safeguard patient privacy and maintain public trust when enabling	Not specified. The paper focuses on governance of real-world health data in general.	Agrees with other studies. It emphasizes the public benefits of secondary health data use while stressing the importance of responsible data governance to protect privacy and maintain trust. It contributes an updated, accessible overview of

			secondary use of health data for public benefit.		access to data for secondary uses through trusted research environments is discuused.		the current UK regulatory landscape and examples to promote progress in enabling access while mitigating risks.
Mone & Shakhlo "Health Data on the Go: Navigating Privacy Concerns with Wearable Technologies"	Discussion of data protection regulations under the EU's GDPR that apply to the use of digital biomarkers in psychiatric research.	Analysis of GDPR articles and principles relevant to digital biomarker research in psychiatry.	Collection and dissemination of sensitive personal data through digital biomarkers in psychiatric research must comply with GDPR principles around consent,	Digital biomarkers (e.g., from smartphones, wearables), but specifics not provided as paper focuses on GDPR considerations rather than	Notes GDPR requirements around security of processing (Article 32) and notification of data breaches (Articles 33-	Digital biomarkers could come from various devices like smartphones and wearables, but specific devices not the focus of the paper.	This study builds on GDPR literature by exploring its application to digital biomarker research in psychiatry. It identifies gaps

			purpose limitation, data minimization, storage limitation, and security.	particular devices/studies .	34), but does not go into technical specifics of how data should be secured.		in current literature regarding positive data protection law considerations for practical solutions in this field.
Belfrage et al. "Trust and digital privacy in healthcare: a cross-sectional descriptive study of trust and attitudes towards uses of electronic health data	Relationship between trust in healthcare system's handling of electronic health data and attitudes towards data use among the general public in Sweden.	Cross-sectional descriptive study using survey responses from 1,645 participants, stratified by age and geographical area.	High levels of trust in healthcare system's handling of electronic health data; trust influences attitudes towards data sharing and perceived risks.	Survey responses from participants.	Not specified in the study.	No device was mentioned; data collected through surveys.	Consistent with previous research indicating high levels of trust in Swedish healthcare system compared to other countries like the UK.

among the general public in Sweden”							
Cai et al. “Continuous quantitative monitoring of physical activity in Parkinson’s disease patients by using wearable devices: a case-control study”	Feasibility of using wearable devices to quantitatively measure daily activity in Parkinson's disease (PD) patients and monitor medication-induced motor fluctuations.	Case-controlled study using smart wrist bracelets to monitor daily movement function of 21 PD patients and 20 healthy volunteers.	Wearable devices can provide continuous, objective monitoring of PD patients' physical activity, detecting differences between PD patients and healthy controls, and motor fluctuations.	Continuous monitoring using smart wrist bracelets.	Not specified in the study.	Smart wrist bracelets (Bong smart bracelet).	Consistent with previous studies exploring the use of wearable devices for monitoring physical activity in PD patients.
Reichmann et al. “The use	Comprehensive review of wearable technology for	Literature review and authors' firsthand	Wearable sensors provide continuous, objective	Wearable sensors (accelerometers, gyroscopes,	Not specifically addressed in the paper, but	Wearable sensors such as the Parkinson's KinetiGraph	The paper aligns with other studies

of wearables for the diagnosis and treatment of Parkinson's disease"	monitoring and managing Parkinson's disease symptoms.	experience with wearable systems (PKG and PDMonitor).	monitoring of PD symptoms, help optimize medication dosing, and enable telemedicine. Most PD patients can use these digital tools.	magnetometers) worn on the body collect movement data. Some systems also include smartphone apps for medication tracking and patient-physician communication .	the authors mention that data from sensors and apps is transferred to a cloud platform accessible by physicians.	(PKG) and PDMonitor, worn on the wrist, waist, and/or legs. Some systems also use smartphone apps.	highlighting the potential of wearable technology for PD management and the importance of telemedicine, especially in light of the COVID-19 pandemic.
Iacobucci "NHS makes urgent appeal for blood donations after cyberattack	Report on a specific ransomware cyberattack affecting multiple	News report based on information from NHS sources and cybersecurity	The cyberattack on Synnovis, an IT provider, disrupted critical hospital services such as blood	Not applicable (no data collection was specified in this news report)	Not applicable (no data security measures discussed in	Not applicable (no devices used for data collection in this news report).	The report is consistent with other news articles and cybersecurity

on London hospitals”	London hospitals and its impact on NHS services.	experts	transfusions and transplant surgeries, forcing trusts to declare incidents and divert patients.		this news report).		analyses discussing the increasing threat of ransomware attacks on healthcare institutions and the need for improved cybersecurity measures.
Poplawska-Domaszewicz et al. (2024) “Parkinson’s Kinetigraph for Wearable Sensor Detection of Clinically	Detection of early morning akinesia in Parkinson’s disease using PKG.	Case series of five Parkinson’s disease patients monitored with PKG.	PKG detected previously unrecognized and untreated early morning akinesia; targeted therapies initiated after PKG	PKG wearable device continuously monitored motor symptoms over 6-7 days.	Not specified in this study.	Parkinson's KinetiGraph (PKG) wearable device.	Aligns with previous studies demonstrating the utility of wearable devices in detecting and monitoring

Unrecognized Early-Morning Akinesia in Parkinson's Disease: A Case Report-Based Observation"			assessments alleviated symptoms.				Parkinson's symptoms.
Atkin et al. (2021) "Perceptions of anonymised data use and awareness of the NHS data opt-out amongst patients, carers and healthcare staff"	Assessment of awareness and perceptions of secondary use of health data in the UK.	Mixed methods: questionnaires and workshops with patients, carers, healthcare staff, and public.	General support for secondary data use by NHS, academic and healthcare-focused commercial organizations; low knowledge of National Data Opt-Out.	Questionnaires and workshops.	Not specified in this study.	Not specified in this study.	Findings align with previous studies showing variable support for health data sharing and low awareness of opt-out systems.
Dominey et al. (2020)	Evaluation of implementing	Prospective service	PKG identified unmet patient	PKG device worn by	PKG data stored in	Parkinson's KinetiGraph	Supports feasibility and

“Introducing the Parkinson’s KinetiGraph into Routine Parkinson’s Disease Care: A 3-Year Single Centre Experience”	the PKG device into routine Parkinson's disease care at a UK centre.	evaluation; PKG data analysis and patient questionnaires.	needs and informed treatment recommendations; generally acceptable to patients but some concerns and areas for improvement.	patients for 6-day periods; questionnaires	proprietary cloud-based system.	(PKG) wrist-worn device.	potential benefits of integrating wearables into clinical care, as shown in other studies.
Wu et al. (2024) “Comparative Efficacy of Commercial Wearables for Circadian Rhythm Home Monitoring from Activity, Heart Rate, and Core	Validating the efficacy of commercial wearable devices for circadian rhythm assessment in real-world home settings.	Two-week longitudinal observational study with 36 participants, collecting activity, temperature, and heart rate data from commercial	Commercial wearable devices can provide accurate and reliable data for circadian rhythm assessment, with acrophase derived from activity, temperature, and	Data collected from a smartwatch (activity and heart rate) and a core body temperature sensor, transmitted to a server during device	Not explicitly discussed in the study.	Galaxy Watch5 smartwatch and CALERAresearch core body temperature sensor.	Findings are consistent with existing research correlating wearable-derived metrics with questionnaire-based evaluations of

Body Temperature”		sensors and comparing them to reference measures.	heart rate data showing strong correlations with reference measures.	charging.			circadian rhythm.
Smuck et al. (2021) “The emerging clinical role of wearables: factors for successful implementation in healthcare”	Identifies factors contributing to the successful implementation of wearable technologies in healthcare by reviewing two effective digital health programs at Kaiser Permanente (diabetes	Qualitative analysis of two real-world digital health interventions by a multi-stakeholder workgroup comprised of 14 experts from diverse backgrounds to identify common features	Seven key features for successful implementation of wearables in healthcare: defining specific problems, incorporating wearables into integrated delivery systems, providing technology support, offering	Data collected through wearable devices (glucose monitors and blood pressure monitors) integrated with electronic health records (EHRs).	Data is transmitted directly to the EHR through secure wireless connections, with access limited to authorized healthcare providers and care teams.	Glucose monitors (Kaiser Permanente program) and blood pressure monitors (Ochsner Health System program) are used to collect patient data.	The identified factors for successful wearable implementation are consistent with other studies emphasizing the importance of integrating wearables into existing healthcare

	management) and Ochsner Health System (hypertension management).	responsible for the success of these programs.	personalized experiences, focusing on end-user needs, aligning with reimbursement models, and engaging clinician champions.				systems, providing user support, and engaging key stakeholders in the adoption process.
Rushdi and Kamal (2012) "Management information systems and the NHS"	Discussed the importance of management information systems (MIS) in supporting decision-making processes within organizations,	Overview of the NHS structure, its modernization program, and the role of IT in improving service delivery; offered recommendation s for future	Highlighted the significance of MIS in the NHS and the potential benefits of interoperable IT systems.	Not specified in this study.	Not specified in this study.	Not specified in this study.	Not explicitly discussed in this study.

	particularly in the context of the NHS.	developments.					
Zhang et al. (2022) "Failing IT infrastructure is undermining safe healthcare in the NHS"	Examined the problem of failing IT infrastructure undermining safe healthcare delivery in the NHS.	Analysis of current state of NHS IT systems and review of relevant literature.	Outdated infrastructure poses significant risks to patient safety and data security; identified key barriers to progress including financial pressures, lack of incentives for NHS leaders, and need for greater clinical engagement.	Not specified in this study.	Highlights the need for improved data security measures and infrastructure.	Not specified in this study.	Aligns with other research on the need for modernizing NHS IT infrastructure and addressing barriers to digital transformation.
Honeyman et al.	Provided an	Reviewed key	The	Not specified	Importance	Not specified in	Findings are

(2016) “A digital NHS?”	overview of the digital health agenda and plans for implementation in the NHS.	government and NHS strategy documents, assessed progress made, and discussed barriers and opportunities.	government's ambitious vision for primary care is hindered by unclear funding and investment, and the need for alignment of technology investments with broader transformation efforts is highlighted.	in this study.	of data security and public trust are mentioned but does not delve deeply into specifics.	this study.	consistent with other assessments of the challenges facing NHS digitization efforts, such as the need for greater funding clarity and stakeholder engagement.
Henshall et al. (2022) “The watch is ticking: A Five-Year plan to	Focuses on the potential of wearable devices to improve health	Proposes a plan to increase wearable adoption through piloting and scaling	Emphasizes the importance of targeting underrepresented groups and addressing data	Not explicitly mentioned in the paper.	Emphasizes the need to address data privacy issues but does not	Mentions wearable devices in general, such as smartwatches and fitness trackers.	Aligns with the general consensus on the potential of wearables to improve

harness wearable health tech”.	outcomes and reduce healthcare costs in the UK.	multiple schemes, such as offering loss-framed incentives.	privacy and security issues to enable widespread adoption of wearables.		provide specific details on data security measures.		health outcomes and the importance of addressing data privacy and security concerns.

Table 3: Comparative Analysis of Studies

2.7 CONCLUSION

In conclusion, the literature review reveals a diversity of opinions on the potential of wearable devices like the PKG to improve remote monitoring and disease management in healthcare. However, there is a growing body of evidence that shows a need for further research on the specific challenges and best practices for integrating these devices with healthcare systems, particularly in terms of data security, governance, and user acceptance. The proposed study aims to address these gaps in the literature by conducting an in-depth case study of the PKG-PDS integration, considering technical, and organizational perspectives to develop recommendations for secure and effective implementation. This research will contribute to the ongoing discussions about the integration of wearable devices in healthcare systems and offer insights into the specific context of the NHS Spine and Parkinson's disease management.

3 METHODOLOGY

3.1 INTRODUCTION

This section outlines the research methodology for critically examining the security of the NHS Spine's PDS and its integration with the PKG wearable device. It provides an overview of the selected research method and approach, data collection methods, data analysis method, and ethical considerations. The methodology is designed to align with the research objectives and ensure a rigorous, comprehensive investigation of the research questions.

3.2 RESEARCH METHOD AND APPROACH

3.2.1 CHOSEN RESEARCH METHOD: QUALITATIVE RESEARCH

Justification for Choosing Qualitative Research

This research employs a qualitative research methodology, which is well-suited for addressing the research questions and objectives. The research objectives are:

1. Provide a concise overview of NHS Spine Core services, investigate the NHS Spine's Personal Demographics Service (PDS), and the key security measures employed to protect patient data within these core services.
2. Analyze the integration of the Parkinson's KinetiGraph wearable device with the PDS, evaluating the security implications and data handling practices specifically related to this integration.
3. Identify critical potential security threats within the PDS and its integration with the Parkinson's KinetiGraph, and propose recommendations to mitigate risks to patient data confidentiality, integrity, and availability.

Qualitative methods are ideal for exploring these complex, multifaceted issues in their real-world context case (Creswell & Poth, 2016). They allow for an in-depth, nuanced understanding of the technical, organizational, and human factors influencing data security in this specific (Korstjens & Moser, 2017). Qualitative research also provides the flexibility to refine the inquiry as new insights emerge during data collection and analysis (Merriam & Tisdell, 2015).

Benefits of Qualitative Approach for this study

A qualitative methodology offers several advantages for assessing the security of the PDS-PKG integration:

1. It enables a thorough examination of multiple layers of data protection, including legal frameworks, policies, technical safeguards, and user practices, revealing how these components work together to mitigate risks and vulnerabilities (Reeves et al., 2013).
2. It can identify potential gaps, inconsistencies, or areas for improvement in existing security measures through document and policy analysis (Flick, 2018).
3. It allows consideration of the social and ethical implications of integrating wearable devices with healthcare systems. Qualitative research can illuminate how stakeholder interests and values are addressed through in-depth analysis of data sharing agreements, consent processes, and patient communication materials (Denzin & Lincoln, 2017).

3.2.2 CHOSEN RESEARCH APPROACH: CASE STUDY

Rationale for Selecting Case Study Approach

A qualitative case study approach was chosen as the most suitable research strategy for this dissertation, given the research questions and objectives. According to Yin (2018), a case study is "an empirical method that investigates a contemporary phenomenon (the 'case') in depth and within its real-world context" (p. 15). This method aligns well with the aim of examining the integration of the NHS Spine's PDS with the PKG wearable device. The case being studied in this research is the integration of the PKG wearable device with the NHS Spine's Personal Demographics Service (PDS).

Case studies are ideal for comprehending complex social phenomena requiring holistic, in-depth investigation (Zainal, 2007), such as the integration of wearable devices with healthcare data systems like the PDS, which involves technological, security, ethical, and governance considerations (Priya, 2020). Additionally, case studies are appropriate when the phenomenon's boundaries and its context are not clearly defined (Yin, 2018), as seen with the security implications of the PDS-PKG integration, which are intertwined with NHS data governance, patient privacy rights, and the evolving landscape of wearable healthcare technologies

3.2.3 ALTERNATIVE RESEARCH METHODS AND APPROACHES

A quantitative method, which analyzes numerical data to test hypotheses and identify statistical relationships (Creswell & Creswell, 2017), could involve gathering data on security incidents or surveying healthcare professionals about PDS security risks. Although quantitative data might provide valuable insights, it may lack the detailed, contextual understanding of complex security, ethical, and governance issues that a qualitative case study offers. Hence, while quantitative methods have advantages, they may not be as effective for exploring the nuanced aspects of PKG integration in healthcare settings..

A mixed methods approach integrates qualitative and quantitative research components (Tashakkori & Creswell, 2007), potentially involving qualitative analysis of PDS and PKG documentation alongside quantitative data collection on system usage, data breaches, or user satisfaction. While this approach can provide a more comprehensive understanding, the exploratory nature of this study and its emphasis on analyzing existing documents and policies suggest that the current qualitative methodology is sufficient to effectively address the research objectives.

A qualitative case study approach was chosen for its alignment with the exploratory nature of the research questions, focus on the specific real-world context of the PDS-PKG integration, and need for an in-depth understanding of the security implications. This method allows for a comprehensive investigation of data handling practices, potential threats, and opportunities for enhancement by analyzing relevant documents, policies, and technical specifications.

3.3 DATA COLLECTION METHODS

Document analysis and literature reviews were selected for this study to obtain detailed information on the technical, policy, and security aspects of integrating the PKG with the NHS Spine's PDS, thus avoiding ethical and logistical issues of directly accessing sensitive healthcare systems and data.

3.3.1 DOCUMENT ANALYSIS

This study on the NHS Spine's PDS and its integration with the PKG will employ document analysis to examine pertinent policies, procedures, and technical specifications, avoiding direct human participation. Documents analyzed will include NHS documentation (policy statements and data-sharing agreements), device manufacturer materials (technical manuals

and product descriptions), and government and regulatory documents (data protection legislation and frameworks).

Inclusion criteria will prioritize documents directly relevant to research objectives, focusing on technical functionalities, data handling, and security measures, while also considering authenticity, reliability, and timeliness.

Document analysis is warranted as it allows detailed examination of policies and protocols governing the PDS and PKG device, minimizing risks and disruptions from accessing sensitive health data. This method supports a rigorous and ethically sound investigation by leveraging existing documentary sources (Morgan, 2022).

3.3.1.1 PKG AND PDS INTEGRATION PROCESS

To provide context for the document analysis, it's important to understand the process of integrating the PKG with the NHS Spine's Personal Demographics Service (PDS). Figure 8 illustrates this integration process:

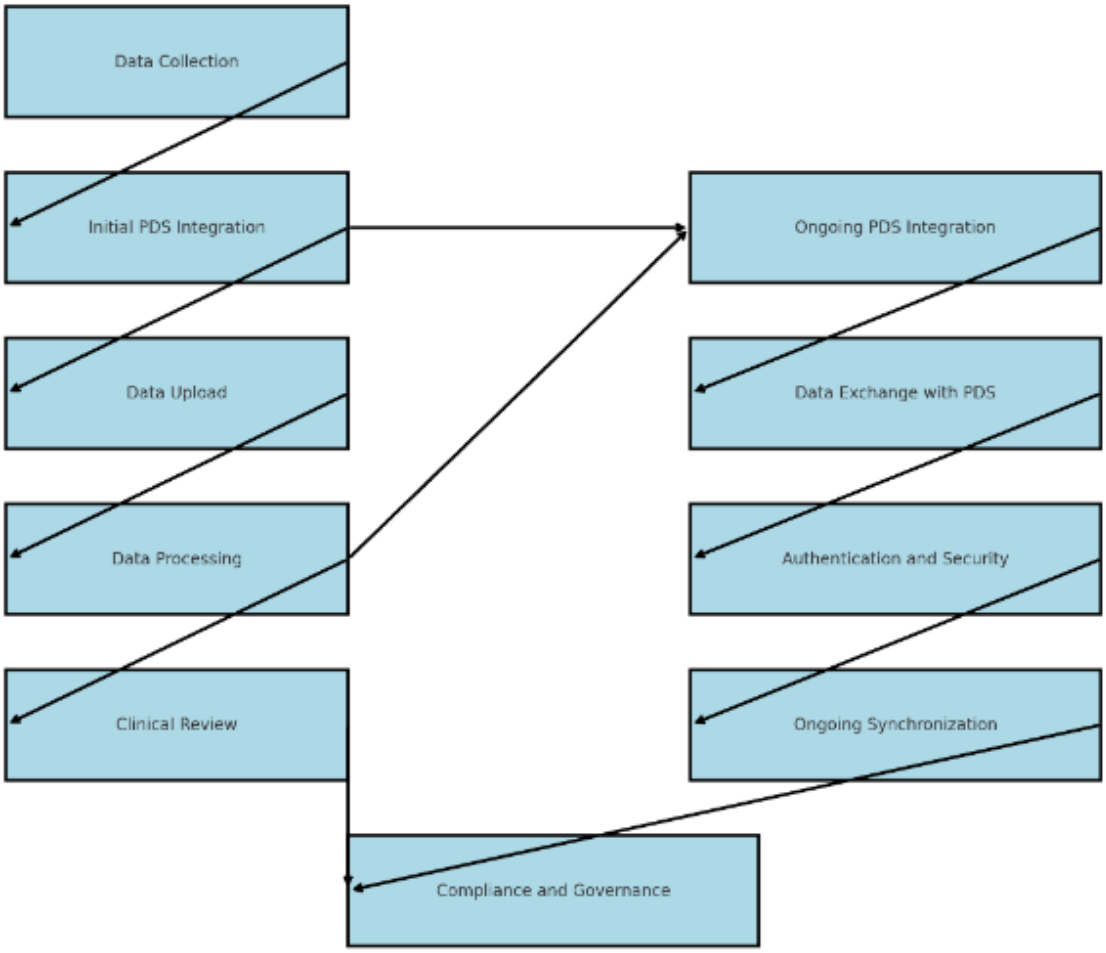

Figure 8: PDS Integration Process Flow

The figure illustrates the integration between a PKG system and the NHS Spine's PDS, detailing steps for collecting, processing, and utilizing patient movement data while maintaining accurate demographic information. Data collection starts with a wearable PKG device, followed by initial PDS integration to verify identity. The data is then uploaded to the HCP portal, processed, and reviewed clinically. Simultaneously, ongoing PDS integration involves data exchange, authentication, and synchronization to ensure accuracy. The figure highlights the importance of compliance and governance overseeing the workflow. This integration efficiently manages patient movement data, ensuring accuracy and security through consistent interaction with the NHS Spine's PDS system. (For an in-depth discussion, see Chapter 5.5.2).

3.3.2 LITERATURE REVIEW

This study will review literature to examine the security implications of integrating the PKG wearable with the NHS Spine's PDS, focusing on potential threats to confidentiality, integrity, and availability. It will cover peer-reviewed articles, conference papers, government reports, and industry publications to provide research context and background. This review will complement the document analysis method, which will examine NHS documentation, device manufacturer materials, and regulatory frameworks. Together, these methods will offer a comprehensive understanding of the current landscape and potential security challenges in integrating PKG wearable devices with NHS systems (Paez, 2017). The literature search strategy is illustrated in Table below.

Category	Details
Search Terms	- "NHS Spine" - "Personal Demographics Service" - "Parkinson's KinetiGraph" - "wearable devices" - "healthcare data security" - "cybersecurity threats" - "data confidentiality" - "data integrity" - "data availability"
Boolean Operators	AND, OR NOT
Example Search Strings	- ("NHS Spine" OR "Personal Demographics Service") AND ("wearable devices" OR "Parkinson's KinetiGraph") - "healthcare data security" AND ("cybersecurity threats" OR "data confidentiality" OR "data integrity" OR "data availability" - ("NHS Spine" OR "Personal Demographics Service") AND "wearable devices" NOT "mobile apps"
Inclusion Criteria	- Published within the last 10 years - English language publications.

	- Peer-reviewed journal articles. - Conference papers - Government reports - Industry publications - Directly relevant to NHS Spine, PDS, or Parkinson's KinetiGraph - Focuses on healthcare data security, wearable devices in healthcare, or cybersecurity threats in healthcare systems
Exclusion Criteria	- Published more than 10 years ago (Unless directly relevant to NHS Spine, PDS, or Parkinson's KinetiGraph). - Non-English language publications - Opinion pieces or editorials - Studies not focused on healthcare data security or wearable devices - Publications not relevant to NHS systems or comparable healthcare data infrastructures.
Databases	- Google Scholar - PubMed - ScienceDirect - IEEE Xplore - Springer Link - thebmj - ACM Digital Library - Web of Science - Scopus -UpToDate
Grey Literature Sources	- NHS Digital website - UK government publications - Cybersecurity industry reports
Document Types	- Academic journal articles - Conference proceedings - Technical reports - White papers - Policy documents - Guidelines and standards

Table 4: Literature Search Strategy

Justification for Literature Review

A comprehensive literature review is essential to establish the current knowledge on the security of the NHS Spine's PDS and its integration with wearable devices like the PKG, identifying security threats, vulnerabilities, and knowledge gaps to inform research questions and ensure innovative contributions (Linnenluecke et al., 2019). The review will also emphasize best practices and recommendations from prior research to guide strategies for enhancing system security, providing a robust foundation for the proposed study.

3.3.3 ALTERNATIVE DATA COLLECTION METHODS

This research employs document analysis and literature review as secondary data collection methods, though other methods like interviews, surveys, and observations could be considered. Interviews involve direct discussions with knowledgeable individuals, such as NHS staff or cybersecurity experts, to examine data handling and security measures. Surveys can gather standardized data from a wider audience but lack the depth of document analysis or interviews. Observations provide insights into behaviors and processes relevant to the research but may require extensive access to NHS facilities, raising privacy and security issues. Interviews, surveys, and observations were not selected due to the appropriateness of document analysis and literature review in meeting research objectives, as well as practical and ethical considerations in studying healthcare system security. These chosen methods enable a thorough investigation of the research questions while minimizing disruptions or risks to the NHS and its stakeholders.

3.4 DATA ANALYSIS METHOD

3.4.1 THEMATIC ANALYSIS

Justification for Selecting Thematic Analysis

Thematic analysis was chosen for its flexibility, detailed account of complex data, and suitability for large textual datasets (Braun & Clarke, 2006; Boyatzis, 1998; Caulfield, 2023). It systematically examines documents to identify key themes and patterns. This method aligns with the research objectives of critically analyzing the NHS Spine's PDS and its

integration with the PKG wearable device, providing insights into data handling practices, security threats, and enhancement opportunities.

Process of Thematic Analysis

As shown in Table5, the thematic analysis will follow the six-phase process outlined by Braun and Clarke (2006):

Phase	Name	Description	Key Activities
1	Familiarization with the data	Immersion in the data to become familiar with its content	- Reading and re-reading the provided documents. - Taking initial notes
2	Generating initial codes	Systematic coding of interesting features across the entire data set	- Identifying interesting aspects in the data items. - Collating data relevant to each code. - Can be done manually or using qualitative data analysis software.
3	Searching for themes	Collating codes into potential themes	- Combining different codes to form overarching theme. - Gathering all data relevant to each potential theme.
4	Reviewing themes	Checking if themes work in relation to coded extracts and the entire data set	- Refining themes. - Combining or separating themes as necessary. - Discarding themes if needed. - Generating a thematic map of the analysis
5	Defining and naming themes	Refining the specifics of each theme and the overall story of the analysis	- Conducting ongoing analysis to refine theme details. - Generating clear definitions for each theme.

			- Creating concise and punchy names for each theme
6	Producing the report	Creating a scholarly report of the analysis	- Selecting vivid, compelling extract examples. - Conducting final analysis of selected extracts. - Relating the analysis back to research questions and literature. - Producing a scholarly report

Table 5: Six-Phase Framework for Thematic Analysis

By following this systematic process, thematic analysis will enable a thorough and rigorous examination of the provided documents, facilitating the identification of key themes and patterns relevant to the research objectives.

3.4.2 ALTERNATIVE DATA ANALYSIS METHODS

This section examines content analysis, discourse analysis, and grounded theory as alternative data analysis methods, but ultimately concludes that thematic analysis is the most suitable choice for this research project as shown in Table 6. Although each method has its strengths, thematic analysis aligns best with the research objectives and the characteristics of the data set. This discussion further justifies the selection of thematic analysis and demonstrates that alternative methods were considered in the research design process.

Data Analysis Method	Description	Strengths	Limitations	Suitability for the Project
Content Analysis	Systematically describes and quantifies phenomena through coding and categorizing textual data to identify patterns and	- Systematic and objective. - Good for identifying patterns and frequencies.	- Focuses more on frequency than deeper meaning. - May miss context and nuance	Less suitable due to focus on frequency rather than deeper meaning and context.

	frequencies of words, phrases, or concepts (Downe-Wamboldt, 1992).		(Vaismoradi et al., 2013).	
Discourse Analysis	Examines how language is used to create and shape social reality, focusing on how language constructs phenomena (Jørgensen & Phillips, 2002).	- Provides insights into how issues are framed and constructed. - Examines language use in depth.	- May not directly address research objectives. - Often involves micro-level. Analysis. - May be too detailed for the scope of the project	Less suitable as it may not directly address the research objectives and could be too detailed
Grounded Theory	Develops theory grounded in systematically gathered and analyzed data through an iterative process of data collection and analysis (Strauss & Corbin, 1997).	- Develops new theory based on data. - Iterative and flexible approach.	- Typically involves collecting original data (Hassan, 2023). - May not be efficient for analyzing existing documents. - Focus on theory development may not align with research objectives	Less suitable due to focus on theory development and typically requiring original data collection.
Thematic Analysis	Identifies, analyzes, and interprets patterns	- Flexible and adaptable.	- May be less rigorous than	Most suitable for the project due to

	of meaning (themes) within qualitative data.	- Allows for systematic examination of data. - Can identify key themes and patterns. - Suitable for analyzing existing documents.	some other methods. - Researcher bias can influence theme identification.	its flexibility, ability to analyze existing documents, and alignment with research objectives.

Table 6: Comparison of Qualitative Data Analysis Methods

Thematic analysis was deemed most appropriate for systematically examining the documents and identifying key themes related to the research objectives.

3.5 ETHICAL CONSIDERATIONS

The proposed research project, evaluating the NHS Spine's PDS and its integration with the PKG, involves analyzing sensitive healthcare data. Ethical implications must be considered, and measures implemented to uphold data privacy, confidentiality, and security, adhering to the UK GDPR and Data Protection Act 2018 (ICO, n.d.; DCMS, 2018).

The research will comply with NHS data governance frameworks, such as the "Five Safes" model and Trusted Research Environments (TREs) (NHS Digital 2023c, NHS Digital 2024e). Responsible disclosure practices will be followed for potential security vulnerabilities, notifying relevant authorities and allowing time for remediation before public disclosure.

Adherence to the academic institution guidelines and professional codes of conduct, along with periodic ethical reviews, will ensure ongoing compliance. By addressing ethical considerations and implementing robust data protection measures, this research aims to respect patient privacy, maintain public trust, and contribute to healthcare data security and the responsible advancement of digital health technologies.

3.6 CONCLUSION

The study adopts a qualitative case study approach to examine the NHS Spine's PDS and its integration with the PKG. Document analysis and literature review are the two data collection methods, with thematic analysis used for data analysis. This methodology allows for a comprehensive investigation of data handling practices, security threats, and enhancement opportunities without directly accessing sensitive health data. The approach aligns with research objectives, enabling an in-depth exploration of healthcare data security in the context of wearable device integration. Ethical considerations are prioritized throughout, adhering to relevant legal frameworks and NHS data governance policies. The Findings chapter will present an objective analysis of the research data, using the previously outlined methodology. It will focus on describing the key themes and sub-themes that emerged from the thematic analysis, directly addressing the research questions.

4 FINDINGS

4.1 INTRODUCTION

The integration of wearable devices with the NHS Spine's PDS presents a critical challenge in safeguarding sensitive patient data, necessitating a comprehensive evaluation of existing security measures. This study aims to critically examine the NHS Spine's PDS and its integration with the PKG wearable device, focusing on the handling of sensitive patient health information, and to identify potential security threats, confidentiality, integrity, and availability issues, ultimately proposing high-level recommendations to enhance the security of these systems. The research objectives are to:

1. Provide a concise overview of NHS Spine Core services, investigate the NHS Spine's Personal Demographics Service (PDS), and the key security measures employed to protect patient data within these core services.
2. Analyze the integration of the Parkinson's KinetiGraph wearable device with the PDS, evaluating the security implications and data handling practices specifically related to this integration.
3. Identify critical potential security threats within the PDS and its integration with the Parkinson's KinetiGraph and propose recommendations to mitigate risks to patient data confidentiality, integrity, and availability.

These objectives address the overarching research questions concerning current security measures, the impact of wearable device integration on data security, and key recommendations for enhancing overall system security.

To address these research aims, a qualitative case study approach was employed, utilizing document analysis and literature review for data collection. Thematic analysis served as the primary method for examining and interpreting the gathered information. The Findings chapter presents a descriptive and objective account of the research results, highlighting the main themes and sub-themes that emerged from the thematic analysis. The chapter is structured around five main themes: NHS Spine and PDS Infrastructure, Security Measures and Data Protection, Integration of Wearable Devices (PKG) and its Impact on Data Security, Security Threats and Vulnerabilities in PDS-PKG Integration and, Recommendations for Enhancing overall security as shown below.

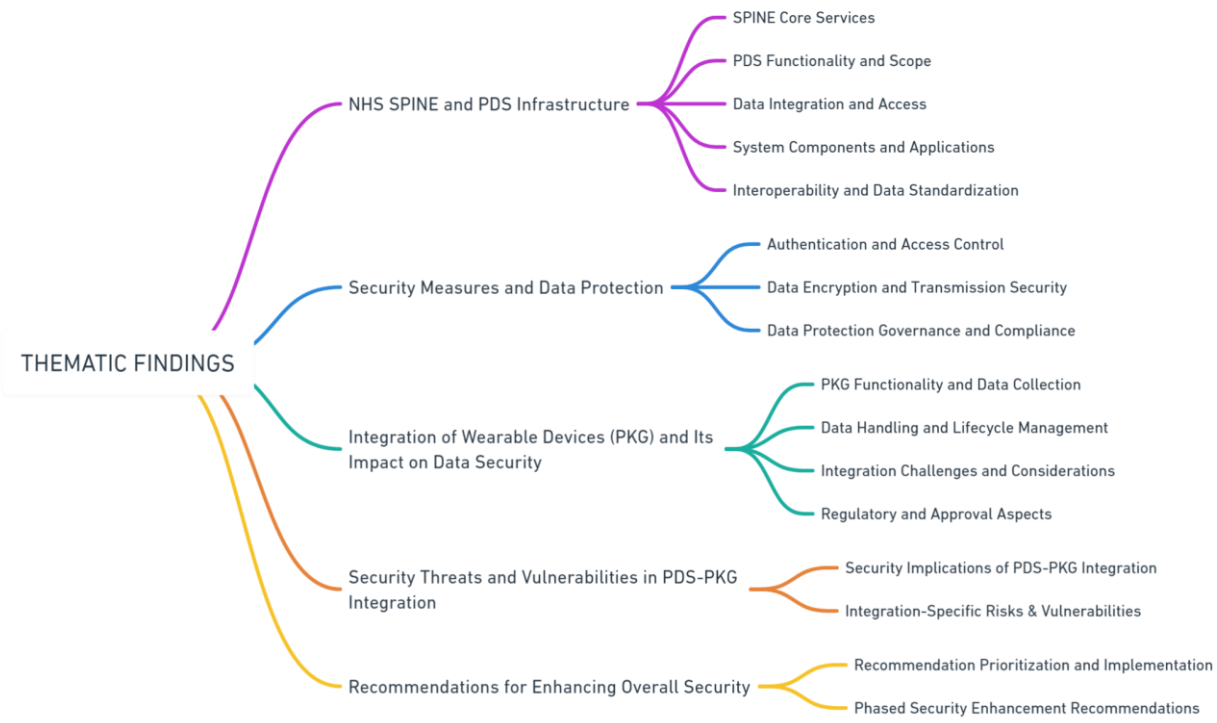

Figure 9: Thematic Map

Each theme explores key aspects of the NHS Spine's PDS and its integration with the PKG with a description and example from the data, providing a holistic view of the security landscape in this evolving healthcare technology ecosystem.

4.2 THEME 1: NHS SPINE AND PDS INFRASTRUCTURE

THEME	DESCRIPTION	EXAMPLES FROM DATA
NHS SPINE AND PDS INFRASTRUCTURE	This theme encompasses the fundamental structure and capabilities of the NHS Spine and its Personal Demographics Service (PDS). It covers the core services provided by the NHS Spine, the functionality and scope of the PDS, data integration and access methods, system components and applications, and system capacity and performance.	"Spine supports the IT infrastructure for health and social care in England… (NHS Digital, 2024j)." "a central IT service called the NHS Spine links your records to your NHS number and makes sure this important information…. (NHS Digital, 2023a)." "Spine allows information held in NHS Digital's national systems to be securely accessed… (NHS Digital, 2024b)."
SUB-THEME		
SPINE CORE SERVICES	This sub-theme focuses on the range of services provided by NHS Spine, highlighting its critical role in healthcare IT infrastructure. It fits into Theme 1 by detailing the specific functionalities that make up the core of the NHS Spine system.	"Spine allows information to be shared securely through national services such as the Electronic Prescription Service, the Personal Demographics Service, the Summary Care Record and the e-Referral Service (NHS Digital, 2024j)."

		"It currently handles over 1.3 billion messages a month and at peak times is processing more than 3,200 messages a second (NHS Digital, 2024j)." "Once your pharmacy implements SCR, you can view SCR on the Spine portal… (NHS Digital, 2024i)."
PDS FUNCTIONALITY AND SCOPE	This sub-theme covers the specific role and capabilities of the Personal Demographics Service within the NHS Spine. It relates to Theme 1 by elaborating on a key component of the NHS Spine infrastructure.	"PDS is the national electronic database of NHS patient demographic data, such as name, address, date of birth and NHS number (NHS Digital, 2021b)." "All patients in PDS have an NHS number which is unique. The 10-digit NHS number is used in England… (NHS Digital, 2024l)." "The information held on the PDS database includes NHS number, name, gender, birth information, address, contact details, registered GP, preferred pharmacy, consent information, related people, such as next of kin, death information (NHS England, 2023a)."

		"It supports quick and accurate identification of a patient, contact and communication with a patient, and linkage of data and records across care settings and information systems (NHS Digital, 2024n)."
DATA INTEGRATION AND ACCESS	This sub-theme addresses how different healthcare systems and professionals interact with and access data through the Spine and PDS. It fits into Theme 1 by highlighting the practical application and utilization of the NHS Spine and PDS infrastructure.	"Most users access Spine through clinical systems or through the Spine portal (NHS Digital, 2024j)." "Spine is for healthcare professionals and health services, including ambulance service, pharmacies, GPs and GP practices, health visitors, child health information services (NHS Digital, 2024j)." "NHS Spine enables healthcare professionals to rapidly access and exchange critical patient information using services such as NCRS, PDS and EPS (NHS Digital, 2024j)." “PDS access methods are: PDS FHIR API, PDS HL7 V3 API … (NHS Digital, 2024n).”

SYSTEM COMPONENTS AND APPLICATIONS	This sub-theme covers the various components and applications that make up the NHS Spine and PDS ecosystem. It relates to Theme 1 by detailing some of the tools and services that comprise the overall infrastructure.	"The demographic batch service (DBS) allows customers to make bulk retrievals of demographic data… (NHS Digital, 2024b)." " The Spine directory service (SDS) database system houses all information associated with the healthcare organizations supporting the NHS. (Gibbs, 2018)."
INTEROPERABILITY AND DATA STANDARDIZATION	This sub-theme focuses on the critical need for seamless data exchange and standardization across NHS Spine and PDS systems. It fits into Theme 1 by addressing key technical aspects of the core infrastructure, including data exchange protocols, standardization efforts, and integration capabilities.	"Interoperability, usability and continual improvement must be central to all future contracting (See Appendix B)". "The digitisation of patient records will lead to major cost, time and space savings standard (See Appendix B)." "This API is the newest and simplest way to integrate healthcare software with PDS. It follows the FHIR (Fast Healthcare Interoperability Resources) standard (NHS Digital, 2024n)." "Simplifying the referral process and supporting national standards of integration and

interoperability (NHS Digital, 2024a)."

Table 7: Findings Theme 1

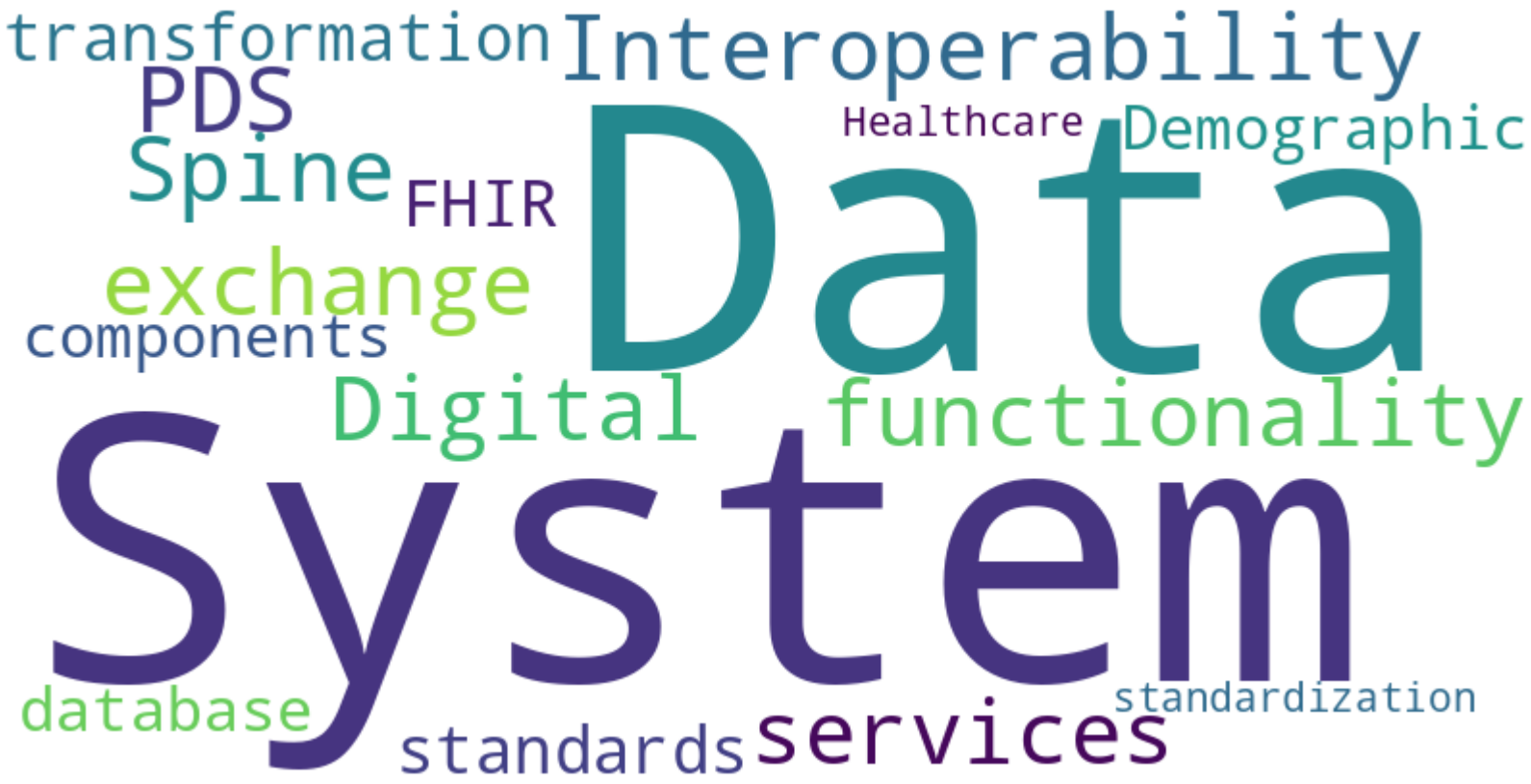

Figure 10: Key phrases and concepts emerged for Theme 1

4.3 THEME 2: SECURITY MEASURES AND DATA PROTECTION

THEME	DESCRIPTION	EXAMPLES FROM DATA
SECURITY MEASURES AND DATA PROTECTION	This theme addresses the critical security measures and data protection practices implemented within the NHS Spine and PDS. It covers authentication and access control methods, data encryption and transmission security, network and infrastructure security, audit and monitoring processes, data protection governance and compliance, and user management and training.	"Spine allows information held in NHS Digital's national systems to be securely accessed (NHS Digital, 2024b)." "The electronic chip in a smartcard stores the unique user identifier (UUID) within the NHS Spine directory, which holds the digital identity information for each smartcard user (NHS England, 2023b)." "The NDG recommends new data security standards for every organisation handling health and social care information. These have been designed to be simple for people to understand and follow... (See Appendix C)."
	SUB-THEME	
AUTHENTICATION AND ACCESS CONTROL	This sub-theme covers the methods and protocols used to verify user identities and manage access to sensitive data within the	"For example, to update a record, healthcare workers must use strong authentication via

	NHS Spine and PDS. It fits into Theme 2 by addressing a fundamental aspect of data security.	the NHS Care Identity Service 2 (NHS Digital, 2024c).” “Each NHS smartcard will have role-based access codes (RBAC) assigned which will allow the user to be electronically authenticated each time they view a patient record (NHS England, 2023b).”
DATA ENCRYPTION AND TRANSMISSION SECURITY	This sub-theme addresses the measures taken to protect data during storage and transmission within the NHS digital infrastructure. It relates to Theme 2 by focusing on the technical aspects of data protection.	“The core tenant of data security is that data must be secure at rest, when stored on a hard disk or memory stick etc., and when in transit, that is being transmitted from one device to another (See Appendix D)." "The different types of secure network connection are Health and Social Care Network (HSCN), Public Services Network (PSN) (NHS Digital, 2024n)." “All health and social care organisations connecting to the NHS Spine must have a valid NHS Public Key Infrastructure (PKI) certificate in place to ensure encrypted patient

		data is transmitted securely (NHS Digital, 2024d)”.
DATA PROTECTION GOVERNANCE AND COMPLIANCE	This sub-theme encompasses the legal, regulatory, and organizational frameworks guiding data protection within the NHS. It fits into Theme 2 by addressing the formal structures that ensure adherence to security standards.	“It is General Data Protection Regulation (GDPR) compliant. (NHS England, 2023a)." "We remind you of your obligations, which are: compliance with Data Security and Protection Toolkit (DSPT). (NHS Digital, 2024m).” "We remind you of your obligations, which are: compliance with NHS Digital cloud storage rules (NHS Digital, 2024n)." “Data protection law enables fair and proportionate data sharing when you approach it in a fair and proportionate way…It provides a framework to help you make decisions about sharing data… (See Appendix E)."

Table 8: Findings Theme 2

Figure 11: Key Phrases and Concepts Emerged for Theme 2

4.4 THEME 3: INTEGRATION OF WEARABLE DEVICES (PKG) AND ITS IMPACT ON DATA SECURITY

THEME	DESCRIPTION	EXAMPLES FROM DATA
INTEGRATION OF WEARABLE DEVICES (PKG) AND ITS IMPACT ON DATA SECURITY	This theme focuses on the integration of wearable devices, particularly the Parkinson's KinetiGraph (PKG), with the NHS Spine and PDS. It covers the functionality and data collection capabilities of the PKG, Wearable Device Data Processing and Integration, data processing and management, and regulatory aspects.	"We hope this decision by NICE will help see use of the PKG and the other remote monitoring devices spread across the NHS (Parkinsons, 2023)." "The PKG® system consists of a wearable recording device, patented algorithms and an HCP portal which provides summaries and advanced analytics to help better understand patients' progression and responsiveness to treatments (PKG Health, 2024)." "The Parkinson's KinetiGraph (PKGTM) consists of a small device worn on the wrist for collecting data over a period of 6-10 days and provides a report for the doctor that shows how motor symptoms and complications of Parkinson's disease (PD) such as slowness of movement, stiffness, tremor and dyskinesias vary throughout the day (PKG Health, 2024)."

SUB-THEME		
PKG FUNCTIONALITY AND DATA COLLECTION	This sub-theme covers the specific capabilities and data collection methods of the Parkinson's KinetiGraph device. It fits into Theme 3 by detailing the core features of the wearable device being integrated.	"The PKG measures bradykinesia, dyskinesia, tremors, motor fluctuations, and immobility, and records when the watch is not being worn. It also monitors movement during sleep (NICE, 2023b)." "The PKGTM also reminds people with PD when it's time to take their prescribed medication whilst helping the patient acknowledging the time the medication was taken... (Dementech, 2022)"
DATA HANDLING AND LIFECYCLE MANAGEMENT	This sub-theme encompasses the comprehensive processes and protocols for managing patient data throughout its entire lifecycle, from initial collection by the PKG wearable device to its integration with NHS systems like PDS FHIR and potential incorporation into Electronic Health Records. The theme relates to Theme 2 by addressing how each stage of data handling contributes to maintaining data integrity, patient privacy, and adherence to NHS standards and regulations.	" Third-party service providers play a crucial role in the storage and processing of personal data in wearable health devices… (Mone & Shakhlo, 2023).” “HCP portal provides one location for managing all patients (PKG Health, 2024).” "It conforms to the FHIR global standard for health care data exchange… (NHS Digital, 2024l).”
INTEGRATION	This sub-theme addresses the technical and	"The device costs £225 per use per patient (NICE,

CHALLENGES AND CONSIDERATIONS	operational challenges of integrating wearable devices like the PKG with existing NHS systems. It relates to Theme 3 by highlighting the complexities involved in implementing new technologies.	2023b)." “The PKG watch is returned by the user to the company (using a prepaid addressed envelope) (NICE, 2023b).” "It can be difficult to assess the symptoms of some people with Parkinson's disease because they can have difficulty communicating, remembering or recording their symptoms ...given that motor fluctuations can vary over time (NICE, 2023b)."
REGULATORY AND APPROVAL ASPECTS	This sub-theme covers the regulatory considerations and approval status of wearable devices like the PKG within the NHS. It relates to Theme 3 by addressing the formal requirements for integrating new technologies.	"They have regulatory approval and are available to the NHS (NICE, 2023b)." "NICE has conditionally recommended 5 remote monitoring devices for Parkinson's for use by the NHS in England... (Parkinsons, 2023)." "In May this year, (NICE) published guidance on their use for remote clinical management of the condition (Neurology Academy, 2021)."

Table 9: Findings Theme 3

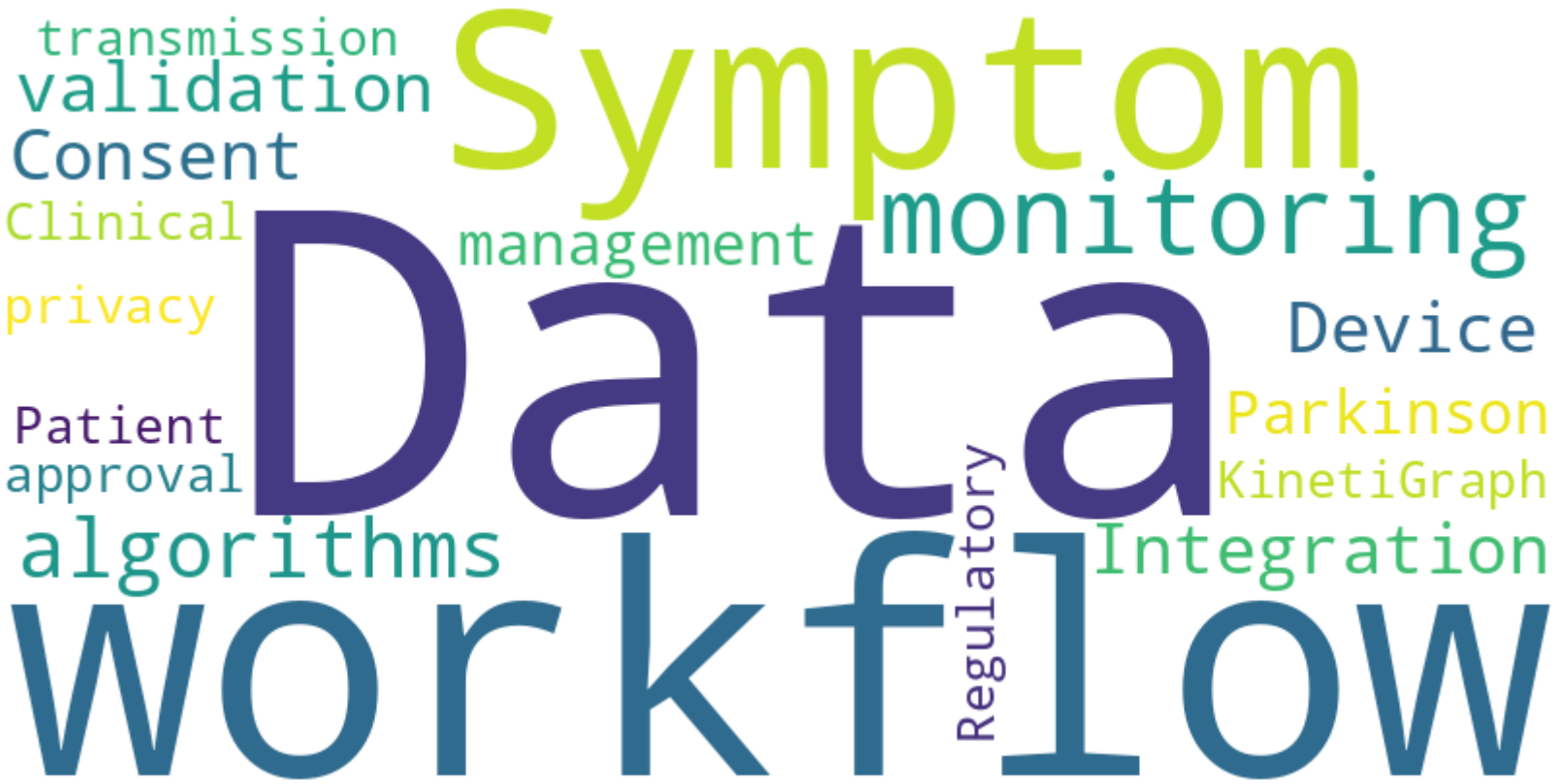

Figure 12: Key Phrases and Concepts Emerged for Theme 3

4.5 THEME 4: SECURITY THREATS AND VULNERABILITIES IN PDS-PKG INTEGRATION

THEME	DESCRIPTION	EXAMPLES FROM DATA
SECURITY THREATS AND VULNERABILITIES IN PDS-PKG INTEGRATION	This theme addresses the potential security threats and vulnerabilities within the NHS Spine, PDS, and their integration with wearable devices like the PKG. It covers various types of cyber threats, system vulnerabilities, risks to data integrity and availability, integration-specific vulnerabilities, user-related vulnerabilities, and systemic and architectural vulnerabilities.	"The 2017 WannaCry attack made clear the need for the NHS to improve cyber security to defend against a future attack... (NHS England, 2023c)." "Common cyber threats that may target NHS staff or patients include junk email or spam, which are irrelevant or unsolicited messages sent via email for the purposes of advertising, phishing, or spreading malware (NHS England, 2023c)."
SUB-THEME		
SECURITY IMPLICATIONS OF PDS-PKG INTEGRATION	This sub-theme focuses on the security implications of connecting the PKG system PDS. It examines how this integration affects data security, highlighting both potential benefits and risks. This fits into Theme 4 by identifying and analyzing specific security	"Connecting the PKG system to the NHS infrastructure expands the potential attack surface (See Appendix F).” "Utilizing the PDS FHIR API for data exchange promotes standardized, secure communication protocols (NHS Digital,

	vulnerabilities and threats that emerge from connecting the PKG system to the NHS Spine's PDS	2024l)." “The integration allows the PKG HCP portal to access and update patient demographic information stored in the PDS (See Appendix F).”
INTEGRATION-SPECIFIC RISKS & VULNERABILITIES	This sub-theme focuses on the specific risks and vulnerabilities that arise from integrating the PKG system with the PDS. It fits into Theme by detailing the potential security issues that could emerge from connecting these two systems.	"The PDS FHIR API's rate limit could lead to availability issues during high-demand periods or if not properly managed (See Appendix F)." “Reliance on PDS for demographic data introduces a potential single point of failure (See Appendix F)." “Vulnerability in the secure storage or transmission of PKG's proprietary algorithms (See Appendix F)."

Table 10: Findings Theme 4

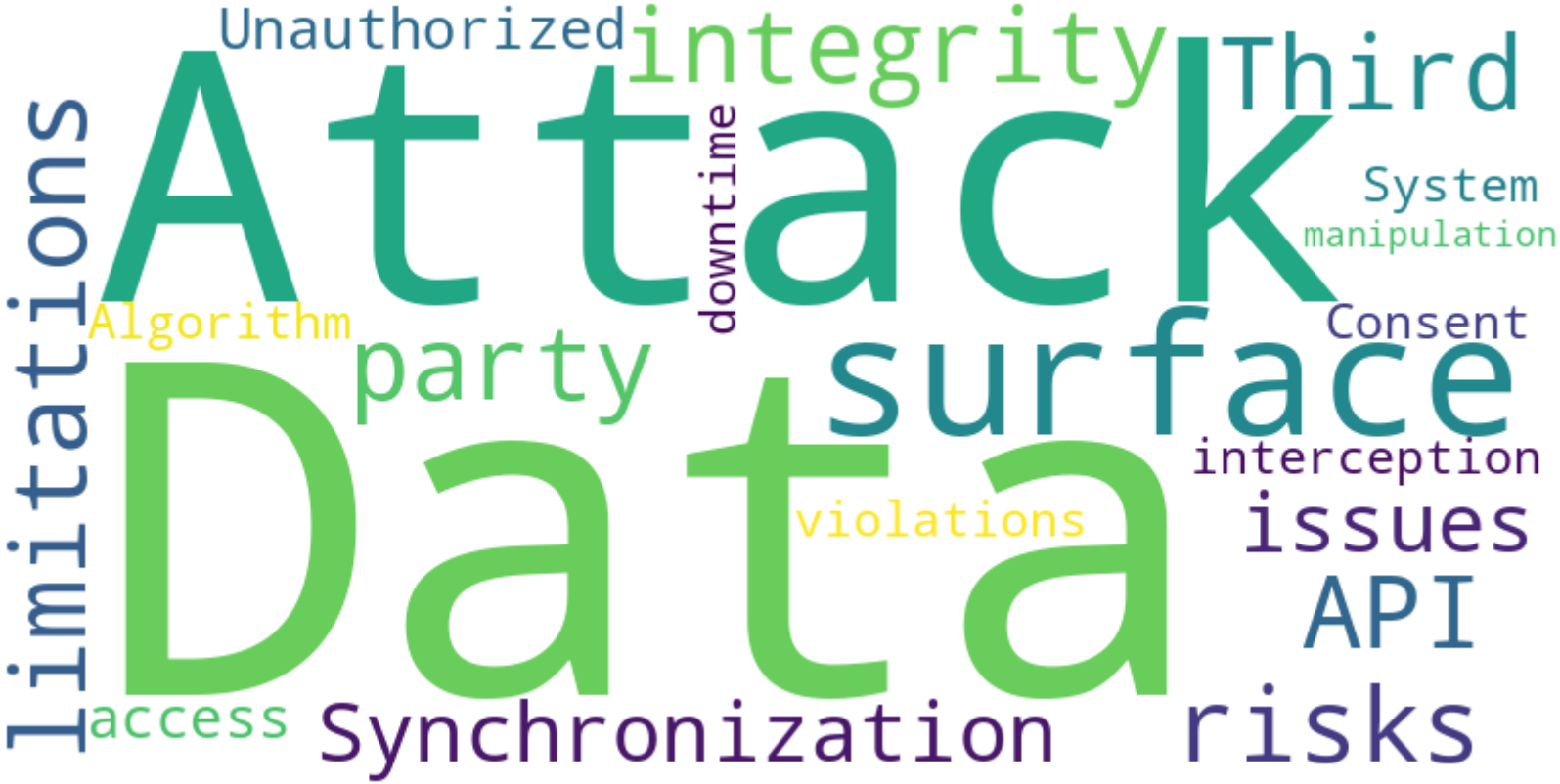

Figure 13: Key Phrases and Concepts Emerged for Theme 4

4.6 THEME 5: RECOMMENDATIONS FOR ENHANCING OVERALL SECURITY

THEME	DESCRIPTION	EXAMPLES FROM DATA
RECOMMENDATIONS FOR ENHANCING OVERALL SECURITY	This theme focuses on the future roadmap for NHS digital systems, including planned enhancements to the PDS and integration of new technologies. It also encompasses recommendations for improving security and data handling practices. The theme covers security enhancement recommendations, Data Enhancement Management and Standardization Recommendations, and strategic approaches and vision.	"Read the PDS roadmap for planned changes to our APIs, including the PDS FHIR API (NHS Digital, 2024l)." "The Secure Data Environment (SDE) is a secure data and research analysis platform. It is part of an interoperable NHS Research Secure Data Environment network (NHS Digital, 2024h)."
SUB-THEME		
RECOMMENDATION PRIORITIZATION AND IMPLEMENTATION	This sub-theme focuses on strategies for prioritizing and implementing security recommendations within the NHS digital ecosystem, including considerations for integrating new technologies like the PKG-PDS. It relates to Theme 5 by providing a structured approach to realizing security enhancements and managing the implementation process across various NHS systems and integrations.	"Include multi-factor authentication for all users accessing the integrated system (See Appendix F)." "Align all security enhancements with NHS Digital's Data Security and Protection Toolkit requirements (See Appendix F)."

		“Ensure that only necessary patient data is transmitted and stored, reducing the potential impact of a data breach (See Appendix F).”
PHASED SECURITY ENHANCEMENT RECOMMENDATIONS	This sub-theme presents an implementation approach for enhancing security measures in a healthcare system. It outlines steps from initial assessment to continuous improvement, prioritizing measures such as robust authentication and secure data transmission. The approach emphasizes stakeholder engagement, including NHS Digital and local NHS trusts, fitting into broader efforts to strengthen cybersecurity in healthcare settings.	“Conduct a comprehensive security assessment of current systems (See Appendix F).” “Begin with "quick wins" - high-priority measures that can be rapidly deployed (See Appendix F).” “Engage with key stakeholders including NHS Digital, local NHS trusts, and the Parkinson's KinetiGraph developers (See Appendix F).”

Table 11: Findings Theme 5

Figure 14: Key Phrases and Concepts Emerged for Theme 5

4.7 CONCLUSION

The thematic analysis of the NHS Spine's Personal Demographics Service (PDS) and its integration with the Parkinson's KinetiGraph (PKG) wearable device revealed several key findings. The NHS Spine and PDS infrastructure emerged as a critical component of the UK's healthcare IT system, providing essential services such as patient demographic data management and facilitating secure information sharing across various

healthcare services. The analysis highlighted robust security measures and data protection practices, including strong authentication mechanisms, encryption protocols, and comprehensive audit processes. However, the integration of wearable devices like the PKG presents opportunities for improved patient care, introduces new challenges and potential vulnerabilities to the system.

The findings also uncovered significant security threats and vulnerabilities, ranging from security implications of PDS-PKG integration and specific risks and vulnerabilities associated with data confidentiality, integrity and availability. Finally, Recommendations for enhancing overall security was focused on prioritization and implementation, and phased security enhancement recommendations for digital transformation in healthcare.

5 DISCUSSION

5.1 INTRODUCTION

This research examines the security implications of integrating the PKG wearable device with the NHS Spine's PDS, focusing on the handling of sensitive patient health information and associated security risks. The study aims to critically evaluate this integration, identify potential threats to data confidentiality, integrity, and availability, and propose recommendations to enhance overall system security. Using a qualitative case study approach, the research employs document analysis and literature review to gather data, followed by thematic analysis to interpret findings. This chapter presents a comprehensive discussion of the research findings, exploring five key themes: NHS Spine and PDS infrastructure, Security Measures and Data Protection, Integration of Wearable Devices and its impact on data security, Security Threats and Vulnerabilities in PDS-PKG integration, and Recommendations for Enhancing Overall Security. Each theme is examined in detail, synthesizing the collected data to address the research questions and objectives.

5.2 OVERVIEW OF KEY FINDINGS

The thematic analysis identified five key themes addressing the research objectives about the security implications of integrating the PKG wearable device with the NHS Spine's PDS.

A prominent theme that emerged was the "NHS Spine and PDS Infrastructure," which provided crucial context for understanding the existing systems and their security measures. This was closely related to the "Security Measures and Data Protection" theme, which highlighted current practices for safeguarding patient data. The analysis also revealed an "Integration of Wearable Devices (PKG) and Its Impact on Data Security" theme, addressing the specific challenges and opportunities presented by incorporating wearable technology. A recurring pattern of "Security Threats and Vulnerabilities in PDS-PKG Integration" emerged, identifying potential risks to data confidentiality, integrity, and availability. Finally, the "Recommendations for Enhancing Overall Security" theme offered strategies to mitigate identified risks and improve system security.

These themes collectively address the research objectives by providing an overview of existing systems, analysing the integration of wearable devices, identifying security threats, and proposing mitigation strategies. The findings suggest that while the integration of

wearable devices like the PKG offers significant benefits for patient care, it also introduces new security challenges that require careful consideration and robust mitigation strategies.

5.3 THEME 1: NHS SPINE AND PDS INFRASTRUCTURE

The NHS Spine and PDS infrastructure forms the backbone of the UK's healthcare IT system, providing essential services for managing patient information and facilitating secure data sharing across various healthcare settings. This theme encompasses the core components, functionality, and integration capabilities of these systems, which are crucial for understanding the context in which the PKG wearable device would be integrated.

The importance of this theme in the context of the study on NHS data security and wearable device integration cannot be overstated. As healthcare systems increasingly rely on digital technologies and data-driven decision-making, the robustness, security, and interoperability of the underlying infrastructure become paramount. Understanding the current state of the NHS Spine and PDS is essential for evaluating the potential risks and opportunities associated with integrating wearable devices like the PKG.

5.3.1 SPINE CORE SERVICES

The findings reveal that the NHS Spine serves as a critical infrastructure, connecting over 28,000 healthcare IT systems across 21,000 organizations. It handles an impressive volume of data, processing over 1.3 billion messages per month and up to 3,200 messages per second during peak times. The table below shows these services.

NHS Spine Core Services	Description
Personal Demographics Service (PDS)	The PDS is a national electronic database of NHS patient demographic details, such as name, address, date of birth, and NHS number. It helps ensure that patient information is consistent and up to date across various NHS services.
Electronic Prescription Service (EPS)	This service allows prescribers (like doctors or nurses) to send prescriptions electronically to a dispenser (like a pharmacy) of the patient's choice. It reduces paper usage, streamlines the prescription process, and helps minimize errors.
Summary Care Record (SCR)	An SCR is an electronic patient record containing essential information about an individual's health, such as allergies, medications, and adverse

	reactions. It's accessible to authorized healthcare staff across England, providing quick access to vital patient information in emergencies or when regular services are unavailable.
e-Referral Service (e-RS)	This service enables healthcare professionals to make electronic referrals to hospitals or clinics for patients. It allows patients to book, change, or cancel appointments online, improving the referral process's efficiency and giving patients more control over their care.

Table 12: Spine Core Services

These findings align with the work of Gibbs (2018), who provided an overview of the NHS Spine network's infrastructure and highlighted its role in connecting various healthcare systems. However, this study offers more recent and specific data on the scale and capacity of the Spine, demonstrating its growth and increased importance in the years since Gibbs' analysis.

The unique contribution of these findings lies in highlighting the sheer scale and complexity of the NHS Spine infrastructure. This underscores the critical need for robust security measures and careful integration processes when considering the addition of new data sources, such as wearable devices like the PKG.

5.3.2 PDS FUNCTIONALITY AND SCOPE

The study highlights the importance of the PDS as the national electronic database for NHS patient demographic information. With over 80 million records, the PDS contains crucial data such as name, address, date of birth, and NHS number. It plays a vital role in supporting patient identification, communication, and data linkage across care settings and information systems to ensure continuity of care and reduce patient misidentification errors. The PDS does not store sensitive data like ethnicity or religion to mitigate the potential impact of data breaches. Authorized healthcare professionals can securely access and share patient demographic information through various interfaces such as the PDS FHIR API, PDS HL7 V3 API, and PDS MESH. The PDS covers records for individuals born in England, Wales, or the Isle of Man; anyone registered for NHS care in these regions; and patients treated by UK Defence Medical Services globally.

These findings align with the emphasis placed on data governance and interoperability in the UK healthcare system as discussed by Jones et al. (2023). The authors highlight the

importance of initiatives like Health Data Research UK (HDR UK) and the Yorkshire Health and Care Record (YHCR) in facilitating secure and responsible use of health data. However, this study provides more specific details on the Personal Demographics Service (PDS) as a national electronic database for patient demographic information, demonstrating concrete progress in establishing standardized data systems within the NHS.

The unique contribution of these findings is the detailed description of the PDS functionality, scope, and data handling practices. While Jones et al. (2023) discuss broader data governance frameworks and initiatives, this study offers insight into a specific, operational system that plays a crucial role in patient identification and data linkage across care settings. This information is vital for understanding the current infrastructure into which wearable devices like the PKG would be integrated, highlighting both the existing capabilities for data sharing and the stringent security measures in place to protect sensitive patient information.

5.3.3 DATA INTEGRATION AND ACCESS

The findings indicate that most users access the Spine through clinical systems or the Spine portal, highlighting the importance of the NHS Spine and PDS in facilitating healthcare data integration across England's National Health Service. The system supports various healthcare professionals and services, including ambulance services, pharmacies, GPs, and child health information services. These users primarily interact with the PDS through the PDS FHIR API, which allows for searching, retrieving, and updating patient information in accordance with global FHIR interoperability standards. System-to-system interfaces enable real-time access to patient demographics across care settings.

New integration methods are being developed to assist system suppliers in designing IT systems that integrate with the Spine or access patient information. Access to PDS is managed through role-based access control (RBAC), ensuring users only access necessary information based on their job functions. Healthcare professionals use NHS smartcards with unique passcodes for authentication, with strong authentication via NHS Care Identity Service 2 (NHS CIS2) required for sensitive operations like record updates.

These findings align with the work of Zhang et al. (2022), who examined the challenges of failing IT infrastructure in the NHS. However, this study provides a more optimistic view, highlighting ongoing efforts to improve integration and access methods. This difference may be due to recent developments and initiatives in the NHS digital transformation strategy.

The unique contribution of these findings is the identification of diverse access points and integration methods for the Spine and PDS. This information is crucial for understanding how wearable devices like the PKG could potentially be integrated into the existing infrastructure, ensuring seamless data flow while maintaining security.

5.3.4 SYSTEM COMPONENTS AND APPLICATIONS

The study identified several key components and applications within the NHS Spine ecosystem, table 13 shows some of the key components.

Key Components	Applications
PDS FHIR (Fast Healthcare Interoperability Resources) API	Provides access to the PDS, search, get, and update patient details
Demographic Batch Service (DBS)	For bulk retrievals of demographic data
Spine Directory Service (SDS)	SDS is a national database of health and social care organizations, workers, and IT systems, providing essential directory services for the Spine ecosystem.
Spine Secure Proxy (SSP)	SSP acts as a broker to control and protect access to GP principal IT systems that expose FHIR-based GP Connect APIs, ensuring secure communications between consumer and provider systems.
Care Identity Service (CIS)	CIS manages NHS smartcard access for over 800,000 users, ensuring secure and auditable access to NHS Spine-enabled health record systems.
Summary Care Record Application (SCRa)	For clinician access to patient information
The NHS App	Provides patients with access to various NHS services and their own health information, including their PDS records.
National Care Records	NCRS allows healthcare professionals to access and update

Service (NCRS)	patient information stored in national systems.
GP2GP	This service allows for the secure transfer of electronic health records between GP practices when patients change doctors.

Table 13: System Components and Applications

These components and services work together to provide a comprehensive suite of digital health services

While previous studies like Honeyman et al. (2016) provided an overview of the NHS digital health agenda, these findings offer a more detailed and up-to-date picture of the specific components and applications within the Spine infrastructure. This granular understanding is essential for identifying potential integration points and security considerations for wearable devices.

The unique contribution of these findings is the detailed mapping of the various components and applications within the NHS Spine ecosystem. This comprehensive view allows for a more nuanced understanding of the potential impact of integrating wearable devices on different aspects of the NHS digital infrastructure.

5.3.5 INTEROPERABILITY AND DATA STANDARDIZATION

The findings underscore the NHS's commitment to enhancing interoperability and data standardization across its digital systems. This includes streamlining referral processes, supporting national integration standards, and promoting modern software development approaches, particularly the adoption of FHIR APIs. These APIs adhere to global healthcare data exchange standards while incorporating UK-specific extensions to meet NHS needs.

The study revealed that the NHS is striving to make data readily available to support patient choice, clinical pathways, and health and social care planning. Key initiatives include:

1. Using the NHS Number as a unique patient identifier
2. Standardizing core demographic data elements
3. Implementing API-based integration
4. Utilizing the centralized Spine Directory Service for healthcare organization and staff information
5. Adhering to data minimization principles

6. Establishing consistent data sharing protocols
7. Standardizing authentication and access control mechanisms

These efforts contribute to the NHS's comprehensive interoperability and standardization strategy, which is crucial for creating a connected, efficient healthcare system that enhances care coordination, reduces errors, and enables advanced analytics and research. However, the findings indicate that achieving true interoperability remains an ongoing process requiring sustained investment, governance, and adaptation to emerging technologies and evolving healthcare needs.

These findings align with the recommendations of Rushdi and Kamal (2012), who emphasized the importance of investing in interoperable IT systems within the NHS. However, this study provides evidence of concrete steps being taken towards achieving this goal, demonstrating progress in the years since their research.

The unique contribution of these findings is the identification of ongoing efforts and specific initiatives aimed at improving interoperability and data standardization within the NHS. This information is crucial for understanding the evolving landscape into which wearable devices like the PKG would be integrated, highlighting both opportunities and potential challenges.

5.4 THEME 2: SECURITY MEASURES AND DATA PROTECTION

The Security Measures and Data Protection theme is central to understanding the safeguards in place to protect sensitive patient information within the NHS Spine and PDS. This theme encompasses various aspects of data security, from access controls to data encryption, governance and frameworks. In the context of integrating wearable devices like the PKG, these security measures are crucial for maintaining the confidentiality, integrity, and availability of patient data.

The importance of this theme cannot be overstated, especially given the increasing cyber threats to healthcare systems and the sensitive nature of the data involved. As the NHS moves towards greater digitization and integration of new technologies, understanding and enhancing these security measures becomes paramount for protecting patient privacy and maintaining public trust.

5.4.1 AUTHENTICATION AND ACCESS CONTROL

These findings reveal the NHS Spine and Personal Demographics Service (PDS) employ several authentication and access control system with multiple security layers. This ensures that only authorized individuals can access sensitive patient data, with access restricted based on roles and responsibilities. The system utilizes two-factor authentication, combining NHS smartcards with unique passcodes, and implements role-based access control (RBAC) to limit user access to necessary information.

The table below illustrate some key authentication methods employed.

Methods	Description	Application
NHS Smartcards	Used in conjunction with a personal passcode, contain a unique user identifier (UUID) stored on the electronic chip, provide secure and auditable access, issued after identity verification.	Primary method of authentication for healthcare professionals accessing NHS Spine-enabled health record systems.
Care Identity Service (CIS)	Manages NHS smartcard access for over 800,000 users, controls access to data in national systems through Role-Based Access Control (RBAC).	Manages authentication and access control for a large number of NHS users.
NHS Care Identity Service 2 (CIS2)	Provides strong authentication for accessing and updating records, offers modern alternatives to smartcards, including biometric authentication, security keys, Windows Hello for Business, Microsoft Authenticator.	Enhances authentication options for accessing and updating records, moving beyond traditional smartcards.
Public Key Infrastructure (PKI)	Provides an additional layer of security for authenticating users and encrypting communications.	Enhances security by ensuring secure communication and user authentication through the use of digital certificates and encryption.

Table 14: Key Authentication Methods

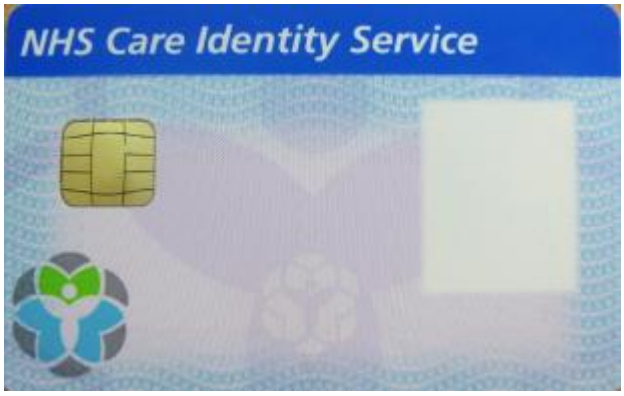

Figure 15: Example of an NHS Smartcard

The Figure below shows how Health and care professionals in England utilize a secure authentication system called NHS CIS2 Authentication to gain access to national clinical information systems. This service introduces alternative authentication methods beyond smartcards, including options such as iPads, security keys, Windows Hello for Business, and Microsoft Authenticator.

Figure 16: NHS CIS2 Authentication

5.4.1.1 ACCESS CONTROL MECHANISMS

This table below outlines key access control mechanisms used in the NHS Spine PDS, describing their functions and applications in managing user access to patient information within the healthcare system.

Mechanism	Description	Application
Role-Based Access Control (RBAC)	Access rights are tailored to specific roles and responsibilities, NHS Digital maintains a national database of roles and permissions, ensures	Cornerstone of access management in the NHS Spine and PDS, ensuring that users only have access to information

	users only have access to necessary information.	relevant to their roles.
Position-Based Access Control (PBAC)	Restricts access based on the user's role within a specific organization and location.	Provides an additional layer of access control by considering the user's position within their organization and location.
Legitimate Relationship Principle	Access is restricted to users who have a legitimate relationship with the patient and their organization	Ensures that only users with a valid reason to access a patient's information are granted access, further enhancing patient privacy.
Spine Directory Service (SDS)	A national database of health and social care organizations, workers, and IT systems, plays a crucial role in managing access rights.	Acts as a central repository for managing access rights across the NHS, enabling effective implementation of access control mechanisms like RBAC and PBAC.

Table 15: Access Control Mechanisms

These findings align with the recommendations of Gandarilla (2018), who emphasized the importance of robust authentication methods and role-based access controls in safeguarding sensitive health data. However, this study provides more detailed information on the specific implementation of these measures within the NHS context.

The unique contribution of these findings lies in highlighting the multi-layered approach to authentication and access control within the NHS. This comprehensive system provides a strong foundation for securing patient data, which is crucial when considering the integration of new data sources like wearable devices.

5.4.2 DATA ENCRYPTION AND TRANSMISSION SECURITY

This study found that the NHS employs several encryption and security measures to protect sensitive patient data, critical for maintaining privacy and trust. Key measures utilized by the NHS include:

1. End-to-end encryption for data transmitted across the NHS Spine, connecting over 28,000 healthcare IT systems, ensuring intercepted data remains unreadable without proper decryption keys.
2. Mandatory use of TLS 1.3 or higher for all API communications, including integrations like the Personal Demographics Service (PDS) FHIR API.
3. Public Key Infrastructure (PKI) for enhanced authentication and encryption, restricting Spine and service access to authorized users with valid digital certificates.
4. Limited access to sensitive NHS systems and data via secure networks, such as the Health and Social Care Network (HSCN), providing a private, secure environment for healthcare data transmission.
5. Implementation of the Spine Secure Proxy (SSP) for certain integrations, like GP Connect APIs, acting as a broker to control and protect access to GP principal IT systems exposing FHIR-based APIs.

These measures collectively aim to ensure protection of patient data both at rest and in transit across the NHS's digital infrastructure.

These findings are consistent with the recommendations of Goswami et al. (2023), they stressed the importance of encryption in securing healthcare big data. However, this study provides specific details on the implementation of these security measures within the NHS infrastructure, offering a more contextualized understanding.

The unique contribution of these findings is the demonstration of a comprehensive approach to data encryption and transmission security within the NHS. This is particularly relevant when considering the integration of wearable devices, as it highlights the need for consistent encryption standards across all data sources.

5.4.3 DATA PROTECTION GOVERNANCE AND COMPLIANCE

The findings reveal a strong emphasis on data Protection Governance and Compliance in the NHS encompasses key regulations, governance structures, and compliance mechanisms. It ensures proper handling of sensitive health data, adhering to laws like UK GDPR and utilizing structures such as Caldicott Guardians.

This table outlines key regulatory frameworks governing data protection and privacy in the UK's National Health Service (NHS). It covers four main elements:

Regulatory Framework	Description
Data Protection Act 2018 and UK GDPR	The NHS must comply with the Data Protection Act 2018, which implements the UK General Data Protection Regulation (GDPR). This legislation sets out key principles for processing personal data, including lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, integrity, and confidentiality.
Health and Social Care Act 2012	This act established NHS Digital (now part of NHS England) as the safe haven for health and care data in England. It provides the legal basis for collecting and processing health and care data at a national level.
Common Law Duty of Confidentiality	This longstanding legal principle requires that information shared in confidence is only disclosed with the individual's permission or when there is an overriding public interest.
National Data Guardian Framework	The role of the National Data Guardian, established by the Health and Social Care (National Data Guardian) Act 2018, provides independent advice and guidance on data sharing and confidentiality in health and social care.

Table 16: Key Regulatory Frameworks

These frameworks collectively ensure the proper handling, protection, and use of health-related data within the NHS system.

This table describes the key governance structures for data protection within the UK's National Health Service (NHS).

Governance Structure	Description
NHS England Data Governance	NHS England, as the primary data controller for national health data, has implemented robust governance structures to oversee data protection:

	- The board, including independent non-executive members, oversees data functions and protection. - The Chief Delivery Officer acts as Senior Information Risk Officer. - Separate directorates handle information governance and data use, ensuring independence.
Caldicott Guardians	Each NHS organization must appoint a Caldicott Guardian, a senior person responsible for protecting patient confidentiality and ensuring appropriate information sharing
Data Access Committees	These committees review and approve requests for access to NHS data, ensuring that data is only shared for appropriate purposes and with adequate safeguards in place.
National Data Opt-Out	This system allows patients to choose whether their confidential patient information can be used for research and planning purposes beyond their individual care.

Table 17: Key Governance Structures

This structure aims to ensure comprehensive data protection and appropriate use of health information across the NHS system.

This table shows key mechanisms used by the UK's National Health Service (NHS) to implement and maintain data protection standards:

Mechanism	Description
Data Security and Protection Toolkit (DSPT)	The DSPT is an online self-assessment tool that enables organizations to measure and publish their performance against the National Data Guardian's data security standards. NHS organizations must complete this assessment annually.
Data Protection Impact Assessments (DPIAs)	DPIAs are mandatory for high-risk data processing activities. They help organizations identify and minimize data protection risks
Information Governance Audits	Regular audits are conducted to ensure compliance with data protection standards and identify areas for improvement.
Incident Reporting	There is a clear process for reporting and managing information-related incidents. Breaches must be reported within 72 hours, as

	required by UK GDPR.

Table 18: Compliance Mechanisms

These mechanisms work together to ensure ongoing compliance, risk management, and continuous improvement in data protection practices across NHS organizations.

These findings align with the work of Jones et al. (2023), who highlighted the importance of understanding the data governance landscape in healthcare. Our study provides more specific details on how these governance frameworks are implemented within the NHS, offering a more nuanced understanding of the compliance landscape.

The unique contribution of these findings is the demonstration of a comprehensive approach to data protection governance within the NHS. This is crucial when considering the integration of wearable devices, as it highlights the need for consistent governance practices across all data sources and technologies.

5.5 THEME 3: INTEGRATION OF WEARABLE DEVICES (PKG) AND ITS IMPACT ON DATA SECURITY

The integration of wearable devices, specifically the PKG, into the NHS infrastructure represents a significant advancement in patient care and monitoring. However, this integration also introduces new challenges and considerations for data security. This theme explores the functionality of the PKG, the challenges of integration, data handling and lifecycle management and the regulatory landscape surrounding wearable devices in healthcare.

Understanding the implications of integrating wearable devices like the PKG is crucial for maintaining the integrity and security of the NHS Spine and PDS while leveraging the benefits of these technologies for improved patient care. This theme is central to addressing the research questions regarding the security implications of integrating the PKG with the PDS and identifying potential vulnerabilities and risks.

5.5.1 PKG FUNCTIONALITY AND DATA COLLECTION

The findings show that the PKG is a wearable device designed to continuously monitor and record motor symptoms in patients with Parkinson's disease. Key aspects of its functionality and data collection process include:

5.5.1.1 CORE COMPONENTS

- Wrist-worn PKG watch
- Proprietary algorithms for data analysis
- Healthcare provider (HCP) portal for data review and management
- Patient and caregiver mobile apps

5.5.1.2 DATA COLLECTION

The section shows the different types of data collected by the PKG wearable device.

Data Type	Details
Continuous Monitoring	The PKG watch is typically worn for 6-10 full days, collecting data during activities of daily living and sleep.
Motor Symptom Tracking	The device measures and records: - Bradykinesia - Dyskinesia - Tremors - Motor fluctuations - Immobility
Technical Specifications	- Weight: 35 grams - Sensors: 3-axis accelerometer - Sampling rate: 50 samples per second - Acceleration range: ±4 g - Data storage: Flash memory
Medication Tracking	- Provides medication reminders - Allows patients to record when medication is taken
Sleep Monitoring	- Captures information about daytime sleepiness and night-time disturbances

Table 19:PKG Watch Specifications and Features for PDM

5.5.1.3 DATA PROCESSING AND ANALYSIS

The section describes the data analysis and process using the PKG

Category	Description
Proprietary Algorithms	Raw movement data is translated into bradykinesia and dyskinesia scores using proprietary mathematical algorithms.
Analysis Techniques	- Fuzzy logic - Spectral power analysis of low-frequency accelerometer traces - Data divided into 2-minute epochs for analysis
Report Generation	The system produces a graphical report summarizing collected data, including: - Median, 25%, and 75% percentiles of bradykinesia and dyskinesia scores - Comparison to control group data - Measures like Fluctuations and Dyskinesia Score (FDS), Percent Time in Tremor (PTT), and Percent Time Immobile (PTI)

Table 20: PKG Data processing and Analysis

5.5.1.4 DATA MANAGEMENT AND ACCESS

This section describes the data management and access components of a Parkinson's disease monitoring system.

Data Transfer	After the monitoring period, the PKG watch is returned to Global Kinetics for data extraction and report generation.
HCP Portal	Healthcare professionals can access reports and advanced analytics through a dedicated portal.
Patient Apps	Complementary mobile apps are available for patients and caregivers, offering features such as: - Customized patient education

	- Side effect management - Patient reported outcomes - Titration guidance

Table 21: PKG Data management and access

These findings align with the study by Cai et al. (2017), which explored the use of smart wrist bracelets for monitoring physical activity in Parkinson's disease patients. However, our study provides more detailed information on the specific functionalities of the PKG, particularly its ability to measure a wide range of motor symptoms and its medication reminder feature.

The unique contribution of these findings lies in highlighting the comprehensive nature of the data collected by the PKG. This level of detail is crucial for understanding the potential security implications of integrating such rich, continuous patient data into the NHS systems.

5.5.2 DATA HANDLING AND LIFECYCLE MANAGEMENT

The findings of this study highlight the critical role of the PKG wearable device and PDS FHIR API integration in promoting interoperability within the NHS. By connecting the PKG device with the NHS Spine's PDS, this integration enables efficient patient identification, consent management, and secure data sharing across healthcare organizations.

5.5.2.1 PKG AND PDS INTEGRATION PROCESS (WORKFLOW)

This section describes a comprehensive process for collecting, processing, and analysing patient movement data using a PKG wearable device, while integrating with the NHS Spine's PDS as shown in Table 22.

Steps	Description
Data Collection	- The PKG wearable device is worn by the patient for 6-10 days. - It continuously collects data on movement, including measures of bradykinesia, dyskinesia, tremors, and other motor symptoms. - The device stores this data on its internal flash memory.
Initial PDS Integration	- When the PKG device is initially assigned to a patient, the system integrates with PDS to verify patient identity through their NHS

	number. - This integration uses the NHS Spine's PDS FHIR API, the preferred method for modern NHS systems. - Patient consent is obtained for accessing and potentially updating their PDS information. - This approach allows for efficient and standardized access to patient demographic information.
Data Upload	- After the monitoring period, the PKG Watch is returned to the company (Global Kinetics) by the patient using a prepaid addressed envelope. - The company extracts the data and generates reports.
Data Processing	- The raw data is processed using proprietary algorithms to generate movement scores, detailed reports, and other analytics. - Reports are generated and made available for healthcare professionals to view online.
Clinical Review	- Healthcare professionals access the PKG reports through the HCP portal provided by the PKG system. - They review the data and may make clinical decisions based on the information.
Ongoing PDS Integration	1. The PKG HCP portal would query the PDS FHIR API to retrieve patient demographic information. This integration primarily involves: a) Retrieving patient demographic information from PDS through their NHS number as a primary identifier to ensure accurate patient identification. b) Potentially updating certain patient details in PDS if necessary (e.g., updating contact information if it has changed). c) Using RESTful API calls conforming to FHIR R4 standards. 2. Throughout the process, the system maintains integration with

	PDS to ensure data accuracy. 3. The integration can occur in real-time or as a batch process through DBS, depending on the implementation.
Data Exchange with PDS	- The interaction occurs through a system-to-system interface through the PKG HCP portal, not directly by end-users. - The system implements robust error handling mechanisms to manage discrepancies or issues in patient identification.
Authentication and Security	- Access to PDS requires strong authentication, using NHS Care Identity Service 2. - Data access to PDS by healthcare professionals is ensured through RBAC. - Data exchange occurs over a secure network connection, such as the Health and Social Care Network (HSCN).
Ongoing Synchronization	The PKG HCP portal maintains its own copy of patient records, it regularly synchronizes with PDS to ensure data consistency.
Compliance and Governance	- Throughout this process, the integration complies with NHS Digital standards for data protection and interoperability. - The organization using the PKG system and accessing PDS demonstrates compliance with the Data Security and Protection Toolkit (DSPT).

Table 22: PKG Wearable Device Data Collection and PDS Integration Process

Key aspects of this integration are as follows:

1. This integration does not involve uploading raw PKG data or detailed movement analytics to PDS. PDS is primarily for demographic information, not clinical data.

2. The integration ensures that PKG data is associated with correct patient records and that relevant updates to patient information are reflected across NHS Spine-connected systems.

3. The process maintains the separation of clinical and demographic data systems, crucial for data protection and system integrity.

4. Patient consent is managed and respected throughout the entire process.

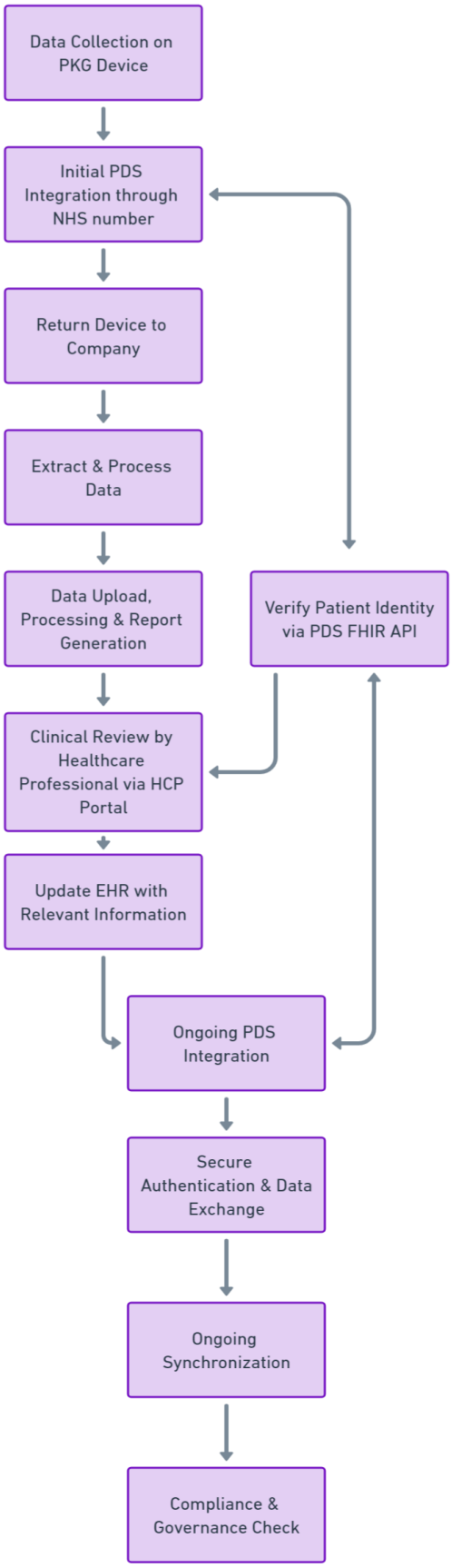

Figure 17: PKG Wearable Device Data Collection and PDS Integration Process

As healthcare systems continue to digitize and connect, integrations like this will become increasingly crucial for delivering coordinated, patient-centered care. By enabling seamless data exchange between devices, systems, and organizations, interoperability initiatives can help improve clinical decision-making, reduce errors, and ultimately enhance patient outcomes.

The findings detail the integration process between the PKG wearable device and the NHS Spine's PDS using FHIR API. This extends Reichmann et al.'s (2023) work on wearable sensors for Parkinson's disease monitoring by providing specific insights on healthcare system integration. The focus on FHIR API integration aligns with Hoeksma's (2019) report on NHS Digital's interoperability efforts, offering practical implementation details for wearable device integration. However, the physical return of devices for data extraction differs from previous studies describing real-time transmission, possibly due to specific security considerations or device limitations.

The unique contribution of these findings is the outlining the specific security measures (e.g., NHS Care Identity Service 2) used in the integration process, providing concrete examples of how security considerations are addressed in practice are addressed in practice.

5.5.3 INTEGRATION CHALLENGES AND CONSIDERATIONS

The integration of wearable devices like the Parkinson's KinetiGraph (PKG) into existing NHS systems presents both significant opportunities and complex challenges. While these devices offer the potential for improved patient monitoring and care, their implementation requires careful consideration of technical, operational, and governance issues.

5.5.3.1 TECHNICAL CHALLENGES

1. Data Interoperability

a) **Issue**: Ensuring seamless data exchange between wearable devices and NHS systems.

b) **Consideration**: The NHS Spine, including services like the Personal Demographics Service (PDS), typically deals with demographic and key clinical data, not raw data from monitoring devices.

2. Security and Encryption

a) **Issue**: Protecting sensitive patient data collected by wearable devices.

b) **Consideration**: NHS systems require end-to-end encryption for data in transit and at rest.

3. Integration with Existing NHS Infrastructure

a) **Challenge**: Connecting wearable devices to the NHS Spine and other national systems through API's like PDS FHIR.

b) **Consideration**: The NHS Spine provides secure data exchange within the NHS **network but may require adaptations for new data types.**

5.5.3.2 **OPERATIONAL CHALLENGES**

1. Data Processing and Analysis

a) **Issue**: Managing and interpreting large volumes of continuous data from wearable devices.

b) **Challenge:** Reliance on patients returning devices for data upload.

c) **Consideration**: Raw data from devices like the PKG is not directly uploaded to NHS systems.

2. Clinical Workflow Integration

a) **Challenge**: Incorporating wearable device data into existing clinical processes.

b) **Issue**: Need for healthcare professionals to access a separate portal for PKG data.

c) **Consideration**: Healthcare professionals need efficient access to relevant patient data without information overload.

3. Training and Support

a) **Issue**: Ensuring healthcare staff can effectively use and interpret wearable device data.

b) **Consideration**: The PKG system, for example, requires specific training for healthcare professionals to interpret reports.

5.5.3.3 GOVERNANCE AND COMPLIANCE CHALLENGES

1. Data Protection and Privacy

a) **Issue**: Ensuring compliance with data protection regulations (e.g., GDPR, Data Protection Act 2018).

b) **Consideration**: NHS systems have strict controls on data access and usage.

2. Clinical Validation and Regulatory Approval

a) **Challenge**: Ensuring wearable devices meet necessary clinical and regulatory standards.

b) **Consideration**: Devices like the PKG have regulatory approvals (FDA cleared, CE marked), but integration into NHS systems may require additional validation.

3. Ethical Considerations

a) **Issue**: Addressing ethical concerns around continuous patient monitoring and data collection.

b) **Consideration**: Patient consent and control over their data are crucial.

These findings are consistent with the work of Smuck et al. (2021), who identified key factors for successful implementation of wearables in healthcare. However, our study provides more specific insights into the challenges of integrating wearables within the NHS context, particularly considering the existing infrastructure and security measures.

The unique contribution of these findings is the identification of NHS-specific integration challenges, which is crucial for developing targeted strategies to address potential security risks associated with wearable device integration.

5.5.4 REGULATORY AND APPROVAL ASPECTS

This study shows that the integration of wearable devices like the PKG into the NHS involves several regulatory considerations and formal approval processes. These are designed to ensure patient safety, data protection, and clinical efficacy.

5.5.4.1 KEY REGULATORY BODIES AND STANDARDS

Key Regulatory Bodies and Standards	Responsibility
National Institute for Health and Care Excellence (NICE)	NICE plays a crucial role in evaluating and recommending new technologies for use in the NHS. For the PKG

	specifically: - NICE has conditionally recommended 5 remote monitoring devices for Parkinson's for use by the NHS in England, including the PKG. - This recommendation is conditional on further data collection regarding how these devices improve health, quality of life, and cost of care. - The guidance was updated in May 2023 to include details of new studies and updates on the UK PKG Registry.
Medicines and Healthcare products Regulatory Agency (MHRA)	MHRA would typically be involved in regulating medical devices like the PKG. They ensure devices meet safety and performance standards.
Information Commissioner's Office (ICO)	The ICO's role is crucial in ensuring compliance with data protection laws, particularly given the sensitive nature of the data collected by wearable devices.
NHS Digital (now part of NHS England)	NHS Digital plays a key role in setting standards for digital health technologies and ensuring their integration with NHS systems.

Table 23: Key UK Regulatory Bodies and Their Responsibilities

5.5.4.2 APPROVAL STATUS AND INTEGRATION REQUIREMENTS

The table below presents an overview of regulatory approvals, NHS integration requirements, and ongoing evaluation processes for the PKG in the UK healthcare system.

Category	**Details**
1. Regulatory Approval	- FDA cleared (United States) - CE marked (European Union) - TGA approved (Australia)
NHS Integration Requirements	
a) Personal Demographics Service (PDS) Integration	- Use PDS FHIR API for patient demographic data - Strong authentication using NHS Care Identity Service 2 (CIS2)

	- Implement role-based access control (RBAC)
b) Data Protection and Security	- Comply with Data Security and Protection Toolkit (DSPT) - Adhere to NHS Digital cloud storage rules (if applicable)
c) Network Security	- Connect via Health and Social Care Network (HSCN) or approved internet-based alternatives
d) Interoperability	- Align with NHS interoperability standards
e) Clinical Safety	- Comply with DCB0129 and DCB0160 clinical safety standards
2. Ongoing Evaluation and Data Collection	- Further data collection on health outcomes, quality of life improvements, and cost-effectiveness - UK PKG Registry for long-term data gathering
3. Ethical Considerations	- Robust patient consent processes for data collection and sharing - Clear protocols for handling sensitive patient data
4. Integration Process	- Digital onboarding process demonstrating valid use case and technical conformance - Possible review by NHS Digital's Information Governance team or Advisory Group for Data (AGD) for novel use cases

Table 24: Regulatory and Integration Requirements for PKG in the UK Healthcare

These findings are consistent with the work of Jones et al. (2023), who emphasized the importance of understanding the regulatory landscape for health data governance. However, our study provides more specific insights into the regulatory considerations for wearable devices within the NHS context.

The unique contribution of these findings is the identification of the dual regulatory challenges – meeting both data protection and medical device regulations – which is crucial for ensuring compliant and secure integration of wearable devices into NHS systems.

5.6 THEME 4: SECURITY THREATS AND VULNERABILITIES IN PDS-PKG INTEGRATION

The integration of the PKG wearable device with the NHS Spine's PDS introduces a new set of security challenges and potential vulnerabilities. This theme explores various security implications, risks and vulnerabilities that arise from this integration. Understanding these risks is crucial for developing effective security measures and ensuring the safe and effective use of wearable devices within the NHS ecosystem.

The importance of this theme cannot be overstated, as it directly addresses the core research questions regarding the identification of potential security threats and vulnerabilities in the PDS-PKG integration. By examining these risks in detail, this study aims to provide insights into the challenges the NHS faces in maintaining data security while adopting innovative technologies for patient care.

5.6.1 SECURITY IMPLICATIONS OF PDS-PKG INTEGRATION

These findings suggests that the integration of the Parkinson's KinetiGraph (PKG) wearable device with the Personal Demographics Service (PDS) presents both opportunities and challenges in terms of data security.

5.6.1.1 DATA SECURITY IMPACT

This section presents an analysis of potential cybersecurity opportunities and challenges arising from integrating the Parkinson's KinetiGraph (PKG) system with NHS infrastructure as shown in Table 25.

Integration Aspect	Potential Opportunities/Challenges
Expanded Attack Surface	Connecting the PKG system to the NHS infrastructure expands the potential attack surface, introducing new potential entry points for malicious actors.
Enhanced Data Access and Patient	- The integration allows the PKG HCP portal to access and update patient demographic information stored in the PDS.

Identification	- This improved access to accurate patient data can enhance the quality of care and reduce the risk of data being associated with the wrong patient. - However, it also increases the potential attack surface for malicious actors.
API Rate Limiting	The PDS FHIR API's rate limit (5 transactions per second per app) could lead to availability issues during high-demand periods if not properly managed.
Heightened Compliance Requirements	Integration with NHS systems require adherence to strict NHS Digital standards, potentially improving overall security practices but also increasing complexity.
Increased Data Transmission	- The integration necessitates more frequent data exchange between systems, potentially increasing exposure points for patient demographic data. - This increases the risk of data interception during transmission, especially if not properly secured.
Standardized Data Exchange	- Utilizing the PDS FHIR API for data exchange promotes standardized, secure communication protocols. - This potentially reduces vulnerabilities associated with non-standard data transfer methods.
Authentication Challenges	- The integration requires robust authentication mechanisms to ensure that only authorized healthcare professionals can access patient data through the PKG HCP portal. - This necessitates the implementation of strong authentication systems like NHS Care Identity Service 2.
Network Dependencies	The integration's reliance on network connectivity (either internet or HSCN) introduces vulnerabilities related to network outages or degradation.

Table 25: Data Security Implications of PKG Integration with NHS Systems

The identification of authentication challenges aligns with the concerns raised by Mone and Shakhlo (2023) regarding the privacy risks associated with wearable devices. However, this

study builds upon their work by illustrating how these challenges are being addressed in a real-world integration scenario, specifically through the use of the NHS Care Identity Service 2.

This unique contribution of these findings is the analysis of the security implications of integrating wearable devices, specifically the PKG, with NHS Spine and PDS. It examines vulnerabilities, risks, and benefits, providing insights for healthcare stakeholders and enhancing understanding of healthcare data security in wearable technology adoption.

5.6.2 INTEGRATION-SPECIFIC RISKS & VULNERABILITIES

This study outlines the potential risks and vulnerabilities associated with integrating the Parkinson's KinetiGraph (PKG) system with the Personal Demographics Service (PDS), categorizing them into confidentiality, integrity, and availability risks. These findings also highlight additional concerns related to consent management, third-party involvement, and regulatory compliance.

5.6.2.1 CONFIDENTIALITY RISKS

This section outlines four major confidentiality risks associated with PDS-PKG integration. Each risk is accompanied by a brief description of potential scenarios or specific concerns related to that risk as shown in Table 26.

Risks	Description
Unauthorized Access	- Increased access points and user roles may elevate the risk of insider threats, where authorized users misuse their access to patient data.
Data Interception	- Potential for interception of data transmissions between the PKG system and PDS if not properly encrypted. - Vulnerability during data transfer, especially if not using secure network connections like HSCN.
Cross-System Vulnerabilities	- A security flaw in either the PKG system or PDS could potentially be exploited to gain unauthorized access to the other system.
Data Leakage	- Risk of unintentional exposure of patient information during data processing or reporting due to inadequate data handling procedures in the PKG system.

Table 26: Key Confidentiality Risks in PKG-PDS Integration

5.6.2.2 INTEGRITY RISKS

This section presents four critical integrity risks associated with PDS-PKG Integration. Each risk is accompanied by a brief explanation of its potential consequences and specific concerns.

Risks	Description
Synchronization Errors	- Regular synchronization between local PKG records and PDS could lead to data inconsistencies if not properly managed. - Potential for inconsistent or outdated patient information.
Data Tampering	- Risk of malicious actors altering patient demographic information during transmission or storage. - This could lead to incorrect patient identification or treatment decisions
Algorithm Manipulation	- Vulnerability in the secure storage or transmission of PKG's proprietary algorithms. - Risk of tampering with data processing algorithms, leading to inaccurate clinical reports.
Data Corruption	- Risk of data corruption during transfer or processing, which could affect the accuracy of patient records.

Table 27: Key Integrity Risks in PKG-PDS Integration

5.6.2.3 AVAILABILITY RISKS

This section outlines four primary availability risks associated with PDS-PKG Integration. Each risk is accompanied by a brief explanation of its potential impact on system availability and patient care.

RISKS	DESCRIPTION
Single Point of Failure	- Reliance on PDS for demographic data introduces a potential single point of failure. - If PDS becomes unavailable, it could impact the PKG system's ability to correctly identify patients or update records.

System Downtime	- Risk of healthcare professionals losing access to critical patient demographic information if the integration between the PKG system and the PDS fails.
API Rate Limiting	- The PDS FHIR API's rate limit could lead to availability issues during high-demand periods or if not properly managed.
Denial of Service (DoS) Attacks	- The integration could potentially make the PKG system a target for denial of service attacks aimed at disrupting access to patient demographic data.

Table 28: Key Availability Risks in PKG-PDS Integration

5.6.2.4 OTHER RELATED RISKS

This table presents three additional significant risks associated with PDS-PKG integration. Each risk is accompanied by a brief explanation of its potential implications and specific areas of concern in the integration process.

RISKS	DESCRIPTION
Consent Management	- Challenge of ensuring proper patient consent for data access and updates. - Risk of unauthorized use of patient data without proper consent.
Third-Party Risk	- Managing security risks associated with Global Kinetics (the company processing PKG data). - Potential data breaches or misuse at the third-party level.
Regulatory Compliance	The integration must comply with both medical device regulations and data protection laws (e.g., GDPR), creating a complex compliance landscape with increased risk of non-compliance.

Table 29: Additional Risks in PKG-PDS Integration

These findings align with the work of Gibbs (2018), who identified potential vulnerabilities in the NHS Spine network, including insider threats and challenges associated with HTTP-based messaging systems. However, this study extends Gibbs' work by providing a more

detailed analysis of the specific risks associated with integrating a wearable device like the Parkinson's KinetiGraph (PKG) with the Personal Demographics Service (PDS).

The unique contribution of these findings is the detailed examination of integration-specific risks and vulnerabilities in the context of wearable devices and centralized healthcare databases. By categorizing risks into confidentiality, integrity, and availability concerns, this study provides a structured framework for evaluating the security implications of similar integrations in the future.

5.7 THEME 5: RECOMMENDATIONS FOR ENHANCING OVERALL SECURITY

The Recommendations for Enhancing Overall Security theme is crucial in addressing the challenges and vulnerabilities identified in Theme 4 regarding the integration of the Parkinson's KinetiGraph (PKG) with the NHS Spine's PDS. This theme encompasses a range of strategies and approaches aimed at bolstering the security of the integrated system while maintaining its effectiveness in patient care.

The importance of this theme cannot be overstated, as it directly addresses the research objective of proposing recommendations to mitigate risks to patient data confidentiality, integrity, and availability. These recommendations form the basis for a comprehensive security strategy that can guide the NHS in the secure integration of wearable devices like the PKG.

5.7.1 RECOMMENDATION PRIORITIZATION AND IMPLEMENTATION

This finding suggests a comprehensive security strategy for integrating the Parkinson's KinetiGraph (PKG) system with the NHS Spine's PDS. Based on the identified security threats and vulnerabilities from Theme 4, key recommendations include implementing robust authentication and access control, ensuring secure data transmission, maintaining compliance with NHS standards, and establishing secure API management, alongside medium-priority measures such as data integrity checks, comprehensive audit trails, and incident response planning. The table below presents a comprehensive list of security recommendations for integrating the PKG with NHS Spine infrastructure such as the PDS.

Recommendations	Priority	Implementation steps

Robust Authentication and Access Control		
- Implement strong authentication using NHS Care Identity Service 2. - Include multi-factor authentication for all users accessing the integrated system. - Implement role-based access control (RBAC) to ensure users only access information necessary for their roles. - Conduct regular access audits and reviews.	High	- Integrate NHS Care Identity Service 2 into the PKG HCP portal. - Define and implement role-based access control policies. - Set up automated access auditing tools.
Secure Data Transmission		
- Ensure all data transmissions between the PKG system and PDS are encrypted using strong, up-to-date protocols. - Use secure network connections like HSCN where possible.	High	- Implement TLS 1.3 or higher for all API communications. - Regularly update encryption protocols to address new vulnerabilities. - Conduct periodic penetration testing of the communication channels.
Compliance and Governance		
- Align all security enhancements with NHS Digital's Data Security and Protection Toolkit requirements. - Ensure compliance with relevant regulations such as the UK Data Protection Act and GDPR.	High	- Review current NHS Digital's Data Security and Protection Toolkit requirements. - Conduct a gap analysis between current security measures and required standards. - Develop and implement a plan to address any identified gaps.
Secure API Management		
- Implement a secure API gateway to manage and monitor all API calls between the PKG system and the PDS. - Include rate limiting, request validation, and detailed logging of all	High	- Select and deploy an API gateway solution compatible with NHS standards. - Configure rate limiting based on PDS FHIR API specifications. - Implement request validation rules and logging mechanisms.

API interactions.		
Integration with Existing NHS Infrastructure (PDS)		
- Ensure all security enhancements are compatible with existing NHS systems and infrastructure. - Leverage existing NHS security frameworks and tools where possible to minimize disruption and reduce costs.	High	- Conduct a thorough assessment of existing NHS security infrastructure. - Identify opportunities for integration with existing tools and frameworks. - Develop an integration plan that minimizes disruption to existing services.
Data Integrity and Synchronization		
- Implement robust synchronization mechanisms to ensure data consistency between the PKG HCP portal and PDS, and to maintain data integrity. - Develop and enforce strict data validation rules for all inputs and outputs. - Implement digital signatures or hash functions for data integrity verification.	Medium	- Develop automated synchronization processes with error handling. - Implement data validation checks before and after synchronization. - Set up alerts for synchronization failures or discrepancies.
Comprehensive Audit Trails and Monitoring		
- Maintain detailed logs of all data access, modifications, and transmissions to detect and investigate potential security breaches. - Implement real-time monitoring of all system activities and data accesses.	Medium	- Implement logging mechanisms for all system interactions. - Set up secure, tamper-evident storage for audit logs. - Develop automated analysis tools to detect suspicious activities.
Regular Security Assessments		
- Conduct frequent security audits and penetration testing to identify and	Medium	- Establish a regular schedule for security assessments.

address vulnerabilities in the integrated system.		- Engage third-party security firms for unbiased evaluations. - Develop a process for quickly addressing identified vulnerabilities.
Data Minimization and Retention Policies		
- Ensure that only necessary patient data is transmitted and stored, reducing the potential impact of a data breach. - Establish and enforce data retention policies.	Medium	- Review and define data requirements for the PKG HCP portal. - Implement data minimization techniques in the integration process. - Establish and enforce data retention policies.
Incident Response Plan		
- Develop and maintain a comprehensive incident response plan to quickly address any security breaches or data loss events.	Medium	- Create an incident response team and define roles. - Develop detailed response procedures for various scenarios. - Conduct regular drills to test the effectiveness of the plan.
Third-Party Risk Management		
- Conduct thorough security assessments of all third-party vendors (e.g., Global Kinetics). - Implement contractual obligations for security standards and practices. - Regularly audit third-party compliance with security requirements.	Medium	- Develop a comprehensive security assessment questionnaire for third-party vendors, focusing on their data handling practices, security controls, and compliance with NHS standards. - Establish a process for regular security audits of vendor systems, including on-site inspections and remote vulnerability assessments. - Implement a vendor risk rating system to prioritize security efforts and allocate resources effectively.
Vendor Management		
- Work closely with the Parkinson's	Medium	- Review and update contracts with PKG

KinetiGraph vendors to ensure their systems meet or exceed NHS security standards. - Establish clear security requirements and SLAs for all third-party vendors involved in the integrated system.		vendors to include specific security requirements. - Establish a process for regular security audits of vendor systems. - Develop clear communication channels for security-related issues with vendors.

Table 30: Security Enhancement Recommendations for PKG-PDS Integration

The emphasis on robust authentication and access control in these findings is consistent with the work of Gandarilla (2018), who stressed the importance of strong authentication methods and role-based access controls in safeguarding sensitive health data. However, this study provides more specific recommendations tailored to the NHS environment, such as the implementation of NHS Care Identity Service 2 and integration with existing NHS security frameworks.

The unique contribution of these findings lies in their comprehensive and integrated approach to security in the context of wearable device integration with NHS systems. While previous studies have addressed various aspects of healthcare data security or wearable technology separately, this research provides a security roadmap that considers authentication, data transmission, compliance, API management, and integration with existing NHS infrastructure.

5.7.2 PHASED SECURITY ENHANCEMENT RECOMMENDATIONS

These finding suggests a phased approach for implementing security enhancements, including assessment, prioritized implementation of measures like robust authentication and secure data transmission, phased execution, continuous improvement, and stakeholder engagement across NHS entities and Parkinson's KinetiGraph developers.
To effectively implement the recommended security measures from the previous sub-theme, the following approach can be proposed:

5.7.2.1 ASSESSMENT AND PLANNING

- Conduct a comprehensive security assessment of current systems.
- Identify gaps between current state and recommended security measures.

- Develop a detailed implementation plan with timelines and resource requirements.

5.7.2.2 PRIORITIZED IMPLEMENTATION

The table below shows the security implementation priorities for the PDS-PKG integration.

High priority (Implement Immediately)	Medium Priority (Implement within 3-6 month)
- Robust Authentication and Access Control - Secure Data Transmission - Compliance and Governance - Secure API Management - Integration with Existing NHS Infrastructure (PDS)	- Data Integrity and Synchronization - Comprehensive Audit Trails and Monitoring - Regular Security Assessments - Data Minimization and Retention Policies - Incident Response Plan - Third-Party Risk Management - Vendor Management

Table 31: Prioritized Implementation for PKG-PDS Integration

5.7.2.3 PHASED EXECUTION

1. Begin with "quick wins" - high-priority measures that can be rapidly deployed.
2. Break down larger initiatives into manageable phases.
3. Implement each recommendation following its specific implementation steps as outlined in the Table above.

5.7.2.4 CONTINUOUS IMPROVEMENT

1. Establish a process for ongoing security review and improvement.
2. Stay informed about emerging threats and new security technologies.
3. Regularly update the security strategy based on new information and changing needs.

5.7.2.5 STAKEHOLDER ENGAGEMENT

1. Engage with key stakeholders including NHS Digital, local NHS trusts, and the Parkinson's KinetiGraph developers.
2. Form a dedicated cross-functional team to oversee the implementation of security enhancements.
3. Ensure clear communication channels are established between all involved parties engage key stakeholders throughout the implementation process.

The phased implementation approach proposed in this study addresses some of the challenges identified by Zhang et al. (2022) regarding outdated IT infrastructure in the NHS. By providing a structured, prioritized plan for security enhancements, this research offers a practical solution to the issues of financial pressures and the need for large-scale digital transformation highlighted in the author's work. The unique contribution here is the presentation of a feasible strategy for gradual improvement that can be adapted to the resource constraints faced by healthcare organizations.

5.8 CONCLUSION

This study investigated three primary questions. The most significant findings are:

1. The NHS Spine and PDS employ robust security measures, including strong authentication (NHS Care Identity Service 2), role-based access control, and end-to-end encryption. However, the integration of wearable devices expands the potential attack surface.
2. The PKG-PDS integration introduces new vulnerabilities, particularly in data transmission, synchronization, and third-party data handling. This integration challenges existing security frameworks designed for traditional healthcare data.
3. Key security risks identified include unauthorized access, data interception, cross-system vulnerabilities, and potential breaches of patient consent in data sharing.
4. High-priority recommendations include implementing enhanced authentication protocols, securing data transmission pathways, ensuring strict compliance with NHS standards, and establishing secure API management.
5. A phased approach to security enhancement is proposed, prioritizing critical measures while addressing medium-priority concerns like comprehensive audit trails and incident response planning.

These findings collectively highlight the complex security landscape created by integrating wearable devices into established healthcare IT systems. While the NHS Spine and PDS have strong foundational security, the integration of devices like the PKG necessitates a re-evaluation and enhancement of existing security protocols. The recommendations provided offer a roadmap for addressing these emerging challenges, emphasizing the need for a dynamic, multi-layered approach to healthcare data security in the era of wearable technology.

6 CONCLUSION

6.1 INTRODUCTION

This chapter will provide a final synthesis of the research on the security implications of integrating the PKG wearable device with the NHS Spine's PDS. It will begin by assessing the fulfillment of research aims and objectives, followed by addressing how the research questions have been answered. The chapter will then present recommendations based on the study's overall findings and suggest directions for future research. Finally, it will conclude with an overall recap of the study's purpose and its significance in the context of healthcare data security and wearable technology integration.

6.2 FULFILMENT OF AIMS AND OBJECTIVES

The research objectives were substantially met through the comprehensive analysis conducted:

1. The study provided a concise overview of NHS Spine Core services and the PDS, detailing key security measures employed to protect patient data. This was achieved through an in-depth examination of the NHS Spine infrastructure and its components, as well as the specific role and capabilities of the PDS.
2. The integration of the PKG wearable device with the PDS was thoroughly analyzed, with particular attention paid to security implications and data handling practices. This analysis revealed both the potential benefits of such integration for patient care and the new security challenges it introduces.
3. Critical potential security threats within the PDS and its integration with the PKG were identified, encompassing risks to data confidentiality, integrity, and availability. Furthermore, the study proposed a set of prioritized recommendations to mitigate these risks, fulfilling the third objective.

6.3 ADDRESSING THE RESEARCH QUESTIONS

The research questions posed at the outset of this study have been comprehensively addressed:

1. Current security measures and data handling practices within the NHS Spine and PDS were found to be robust, including strong authentication mechanisms, role-based

access control, and end-to-end encryption. However, the integration of wearable devices introduces new challenges to these existing security frameworks.

2. The impact of integrating the PKG with the PDS was found to be significant, introducing new vulnerabilities particularly in areas of data transmission, synchronization, and third-party data handling. The study identified specific risks such as unauthorized access, data interception, and potential breaches of patient consent in data sharing.
3. Key recommendations for enhancing overall system security were proposed, prioritizing measures such as implementing enhanced authentication protocols, securing data transmission pathways, ensuring strict compliance with NHS standards, and establishing secure API management. A phased approach to security enhancement was suggested to address both high-priority and medium-priority concerns.

6.4 RESEARCH CONTRIBUTIONS AND FUTURE DIRECTIONS

From a research perspective, this study addresses a critical gap in the literature regarding the security implications of integrating wearable devices with centralized healthcare databases like NHS Spine's PDS. While previous research has explored either wearable technology in healthcare or security in healthcare IT systems, few studies have examined the intersection of these two areas in depth. By focusing on this intersection, the current study provides a foundation for future research in this rapidly evolving field. The identification of specific security risks and vulnerabilities in the PKG-PDS integration serves as a starting point for more targeted investigations into security solutions for wearable healthcare devices.

Furthermore, this study contributes by identifying several key areas for future research:

1. Empirical studies on the real-world implementation of wearable device integration with NHS systems, including quantitative assessments of security incidents and their impacts.
2. In-depth analysis of patient perspectives on the use of wearable devices and the sharing of their health data within integrated healthcare systems.
3. Comparative studies of similar integrations in other healthcare systems globally to identify best practices and potential areas for improvement.

4. Long-term studies on the evolving threat landscape specific to healthcare wearables and their integration with centralized health databases.
5. Research into the development of privacy-enhancing technologies specifically designed for wearable health devices and their integration with healthcare IT systems.

These identified areas for future research provide a roadmap for scholars and practitioners to build upon the findings of this study and further advance the field of healthcare IT security in the context of wearable device integration

6.5 RECOMMENDATIONS

Based on the findings of this study, several key recommendations can be proposed:

1. Implement a multi-layered security approach that addresses both technical and organizational aspects of data protection.
2. Ensure secure data transmission through end-to-end encryption and secure API management.
3. Prioritize the enhancement of authentication and access control mechanisms, particularly in the context of integrating new technologies like wearable devices.
4. Develop and implement comprehensive data governance policies that specifically address the challenges posed by wearable device integration.
5. Maintain strict compliance with NHS standards and regulatory requirements.
6. Establish robust incident response and disaster recovery plans to mitigate the impact of potential security breaches.
7. Invest in ongoing security training and awareness programs for all stakeholders involved in the use and management of integrated healthcare IT systems.
8. Regularly conduct security risk assessments and audits to identify and address emerging vulnerabilities in the integrated system.

6.6 CONCLUSION

Given the critical importance of data security in healthcare, this study sought to evaluate the security implications of integrating the Parkinson's KinetiGraph (PKG) wearable device with the NHS Spine's Personal Demographics Service (PDS). This chapter concluded the study by highlighting the key findings in relation to the research aims and questions. Notably, while current NHS Spine and PDS security measures were found to be robust, the integration of wearable devices introduced new vulnerabilities, particularly in data transmission,

synchronization, and third-party data handling. Based on these findings, actionable recommendations were made, including implementing a multi-layered security approach and enhancing authentication mechanisms. The chapter also identified areas for future research, such as empirical studies on real-world implementations and investigations into privacy-enhancing technologies for wearable devices. This research contributes to bridging the gap between wearable technology in healthcare and security in healthcare IT systems, providing a foundation for future studies in this rapidly evolving field.

7 CRITICAL REFLECTION (EVALUATION)

This chapter critically reflects on the research process and findings. While Chapter 6 summarized the fulfilment of research objectives and addressed research questions, this chapter evaluates the methodology, discusses limitations, and explores broader implications for industry practice.

7.1 LIMITATIONS AND IMPROVEMENTS OF RESEARCH METHODS

This study employed a qualitative case study approach, focusing on document analysis and literature review for data collection. This methodology allowed for a thorough examination of the security implications of integrating the Parkinson's KinetiGraph (PKG) with the NHS Spine's Personal Demographics Service (PDS). However, critical evaluation of the study's limitations and potential improvements is necessary.

A significant limitation was the reliance on secondary data sources. The study analyzed existing documents and literature rather than collecting primary data from stakeholders involved in the PKG-PDS integration. Although this approach offered a comprehensive overview of existing knowledge and policies, it may have limited the study's ability to capture current practices and challenges. Yin (2018) emphasizes that case studies benefit from multiple evidence sources to enhance validity through triangulation.

The lack of a primary data collection meant the study could not directly capture the perspectives of healthcare professionals, IT security experts, and patients affected by the integration of wearable devices with NHS systems. This limitation may impact the findings by not fully reflecting practical challenges in implementing security measures in real-world healthcare settings. Creswell and Poth (2016) argue that qualitative research gains richness from diverse participant viewpoints.

To address these limitations, future research could adopt a mixed-methods approach, combining document analysis with primary data collection methods such as interviews or surveys. Semi-structured interviews with NHS IT security professionals, clinicians using the PKG system, and device manufacturer representatives could provide insights into the practical challenges of maintaining security in the integrated system, aligning with Johnson et al. (2007) on methodological triangulation benefits.

A survey of patients using PKG devices could offer important perspectives on their understanding and concerns about data security, addressing a gap identified in this study. This aligns with the emphasis on patient-centered research in healthcare, as advocated by Greenhalgh et al. (2019).

Replicating this study could include a comparative case study element, examining similar wearable device integrations in other healthcare systems internationally, providing benchmarks and identifying best practices for the NHS context. Goodrick (2014) notes that comparative case studies enhance the generalizability of findings across different contexts.

Additionally, a more systematic and comprehensive review of technical documentation related to the PKG-PDS integration could be conducted. While the current study examined available public documents, an in-depth technical analysis, including system architecture diagrams or security protocols (with appropriate permissions), could provide a more granular understanding of potential vulnerabilities and security measures.

7.2 IMPLICATIONS AND RECOMMENDATIONS FOR INDUSTRY

Despite these limitations, this study has significant implications for integrating wearable devices with centralized healthcare systems.

The findings highlight the complex security landscape created by integrating wearable devices into healthcare IT infrastructures. While the NHS Spine and PDS have robust security measures, integrating devices like the PKG introduces new vulnerabilities. This necessitates a more dynamic approach to security, with healthcare organizations and technology providers constantly evolving their strategies to address emerging threats.

Based on these findings, healthcare technology providers and NHS IT teams should collaborate closely in the early stages of wearable device integration projects. This collaboration should involve comprehensive security assessments considering both individual components and risks arising from their integration, as suggested by Coventry and Branley (2018).

The study underscores the importance of robust authentication and access control mechanisms in integrated healthcare systems. Implementing enhanced protocols, such as the

NHS Care Identity Service 2, implies a need for investment in advanced identity management solutions. Practitioners should conduct regular security audits focusing on access control and authentication in integrated systems, assessing technical controls, user behaviors, and organizational policies (Fernández-Alemán et al., 2013).

Findings related to data transmission and third-party handling risks imply a need for greater scrutiny of data flows in integrated systems. Healthcare organizations and technology providers should develop comprehensive data governance frameworks addressing wearable device integration challenges. These should define data ownership, establish secure transmission protocols, and set requirements for third-party handlers (Azarm-Daigle et al., 2015).

Future research should focus on developing privacy-enhancing technologies for wearable health devices and their integration with healthcare IT systems. This could include advanced encryption methods, secure multi-party computation, or blockchain-based solutions for maintaining data integrity.

Another critical area for research is the long-term impact of wearable device integration on patient trust and engagement with healthcare systems. This aligns with the growing recognition of patient perspectives in healthcare technology adoption (Greenhalgh et al., 2017).

The industry should prioritize research into scalable security solutions accommodating the growing variety of wearable health devices. As the Internet of Medical Things expands, developing flexible yet robust security frameworks will be crucial.

In conclusion, this study provides valuable insights into the security challenges and opportunities presented by wearable device integration with centralized healthcare systems. Addressing the identified limitations and acting on the recommendations can lead to more secure and effective integration, improving patient care while protecting sensitive health information.

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APPENDICES

APPENDIX A – ETHICS CHECKLIST

Sheffield Hallam University

UREC 1 RESEARCH ETHICS REVIEW FOR STUDENT RESEARCH WITH NO HUMAN PARTICIPANTS OR DIRECT COLLECTION OF HUMAN TISSUES, OR BODILY FLUIDS.

All University research is required to undergo ethical scrutiny to comply with UK law. The University Research Ethics Policy (www.shu.ac.uk/research/excellence/ethics-and-integrity/policies) should be consulted before completing this form. The initial questions are there to check that completion of the UREC1 is appropriate for this study. The supervisor will approve the study, but it may also be reviewed by the College Teaching Program Research Ethics Committee (CTPREC) as part of the quality assurance process (additional guidance can be obtained from your College Research Ethics Chair[1]).

The final responsibility for ensuring that ethical research practices are followed rests with the supervisor for student research.

Note that students and staff are responsible for making suitable arrangements to ensure compliance with the General Data Protection Regulations (GDPR), for keeping data secure and if relevant, for keeping the identity of participants anonymous. They are also responsible for following SHU guidelines about data encryption and research data management. Guidance can be found on the SHU Ethics Website www.shu.ac.uk/research/excellence/ethics-and-integrity

Please note that it is mandatory for all students to only store data on their allotted networked F drive space and not on individual hard drives or memory sticks etc.

This form also enables the University and College to keep a record confirming that research conducted has been subjected to ethical scrutiny. Students should retain a copy for inclusion in their research projects, and a copy should be uploaded to the relevant module Blackboard site.

The form must be completed by the student and approved by supervisor and/or module leader (as applicable). In all cases, it should be counter-signed by the supervisor and/or module leader and kept as a record showing that ethical scrutiny has occurred. Students should retain a copy for inclusion in the appendices of their research projects, and a copy should be uploaded to the module Blackboard site for checking.

Please note that it may be necessary to conduct a health and safety risk assessment for the proposed research. Further information can be obtained from the University's Health and Safety Website https://sheffieldhallam.sharepoint.com/sites/3069/SitePages/Risk-Assessment.aspx

ARE YOU COMPLETING THE CORRECT FORM?

[1] College of Social Sciences and Arts – Dr. Antonia Ypsilanti (a.ypsilanti@shu.ac.uk)
College of Business, Technology and Engineering – Dr. Tony Lynn (t.lynn@shu.ac.uk)
College of Health, Wellbeing and Life Sciences – Dr. Nikki Jordan-Mahy (n.jordan-mahy@shu.ac.uk)

UREC 1: Application for Research Ethics Approval for Student Studies with No Human Participants

Ethics Form

APPENDIX B

Technology, infrastructure and dat

Technology, infrastructure and data supporting NHS staff

APPENDIX C

National Data Guardian.pdf

National Data Guardian for Health and Care Review of Data Security, Consent and Opt-Outs

APPENDIX D

General Practice IT Infrastructure Specif

General Practice IT Infrastructure Specification

APPENDIX E

data sharing a code of practice.pdf

Data sharing code of practice

APPENDIX F

C5 Vulnerabilities, Risks and Recomme

Chapter 5 Vulnerabilities, Risks and Recommendations

APPENDIX G

List of abbreviations and their meanings.

ABAC: Attribute-Based Access Control

AGD: Advisory Group for Data

API: Application Programming Interface

CIS: Care Identity Service

CIS2: Care Identity Service 2

DBS: Demographic Batch Service

DCMS: Department for Digital, Culture, Media and Sport

DoS: Denial of Service

DSPT: Data Security and Protection Toolkit

e-RS: e-Referral Service

EPS: Electronic Prescription Service

FDA: Food and Drug Administration

FHIR: Fast Healthcare Interoperability Resources

FDS: Fluctuations and Dyskinesia Score

GDPR: General Data Protection Regulation

GP: General Practitioner

HCP: Healthcare Provider

HDR UK: Health Data Research UK

HIPAA: Health Insurance Portability and Accountability Act

HSCN: Health and Social Care Network

ICO: Information Commissioner's Office

IoT: Internet of Things

IT: Information Technology

MHRA: Medicines and Healthcare products Regulatory Agency

MIS: Management Information Systems

NDG: National Data Guardian

NCRS: National Care Records Service

NHS: National Health Service

NICE: National Institute for Health and Care Excellence

PBAC: Position-Based Access Control

PD: Parkinson's Disease

PDS: Personal Demographics Service

PKG: Parkinson's KinetiGraph

PKI: Public Key Infrastructure

PSN: Public Services Network

PTI: Percent Time Immobile

PTT: Percent Time in Tremor

RBAC: Role-Based Access Control

SCR: Summary Care Record

SCRa: Summary Care Record Application

SDE: Secure Data Environment

SDS: Spine Directory Service

SSP: Spine Secure Proxy

TGA: Therapeutic Goods Administration (Australia)

TLS: Transport Layer Security

TRE: Trusted Research Environment

UK: United Kingdom

UUID: Unique User Identifier

YHCR: Yorkshire Health and Care Record